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(54) **SYSTEMS AND METHODS FOR EXTENDED LONG RANGE COMMUNICATION IN WIRELESS LOCAL AREA NETWORKS (WLANS)**

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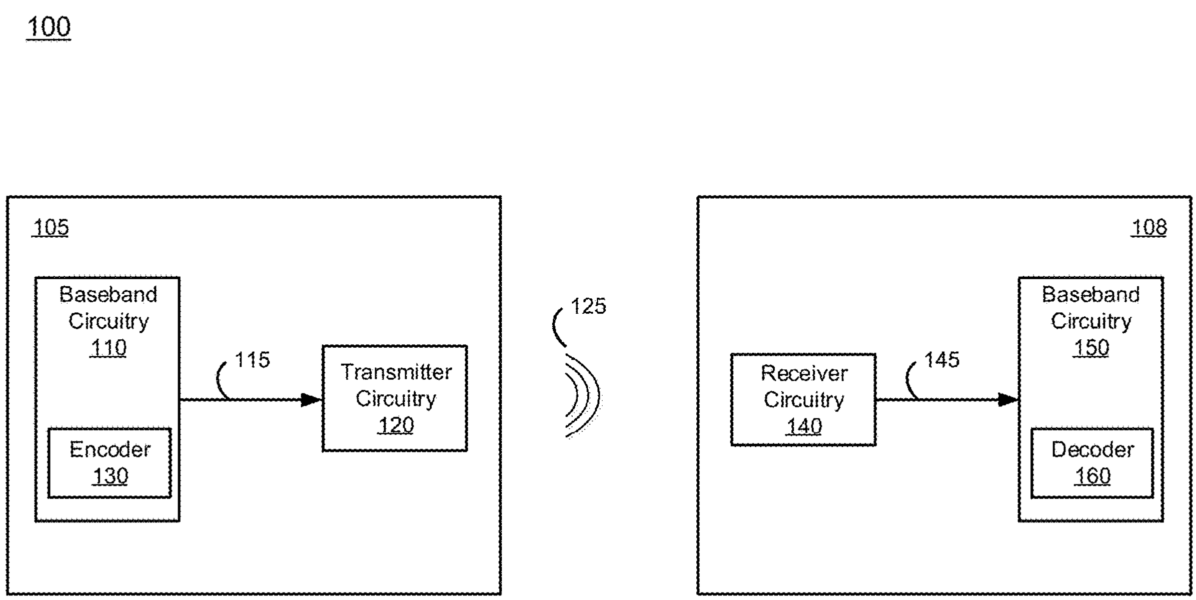
(57) **ABSTRACT**

In some implementations, an apparatus may include a transmitter and one or more processors. The one or more processors may identify a target data rate for transmitting data over a channel with a frequency bandwidth. Based at least on the target data rate, the one or more processors may select a forward error correction (FEC) code, a code rate, a modulation scheme, and a number of resource units (RUs) within the frequency bandwidth, to transmit the data within a range of the target data rate. The one or more processors may encode, by an FEC encoder, the data using the FEC code and the code rate to generate encoded data. The one or more processors may modulate the encoded data using the modulation scheme to generate modulated data. The transmitter may transmit the modulated data using the number of RUs.



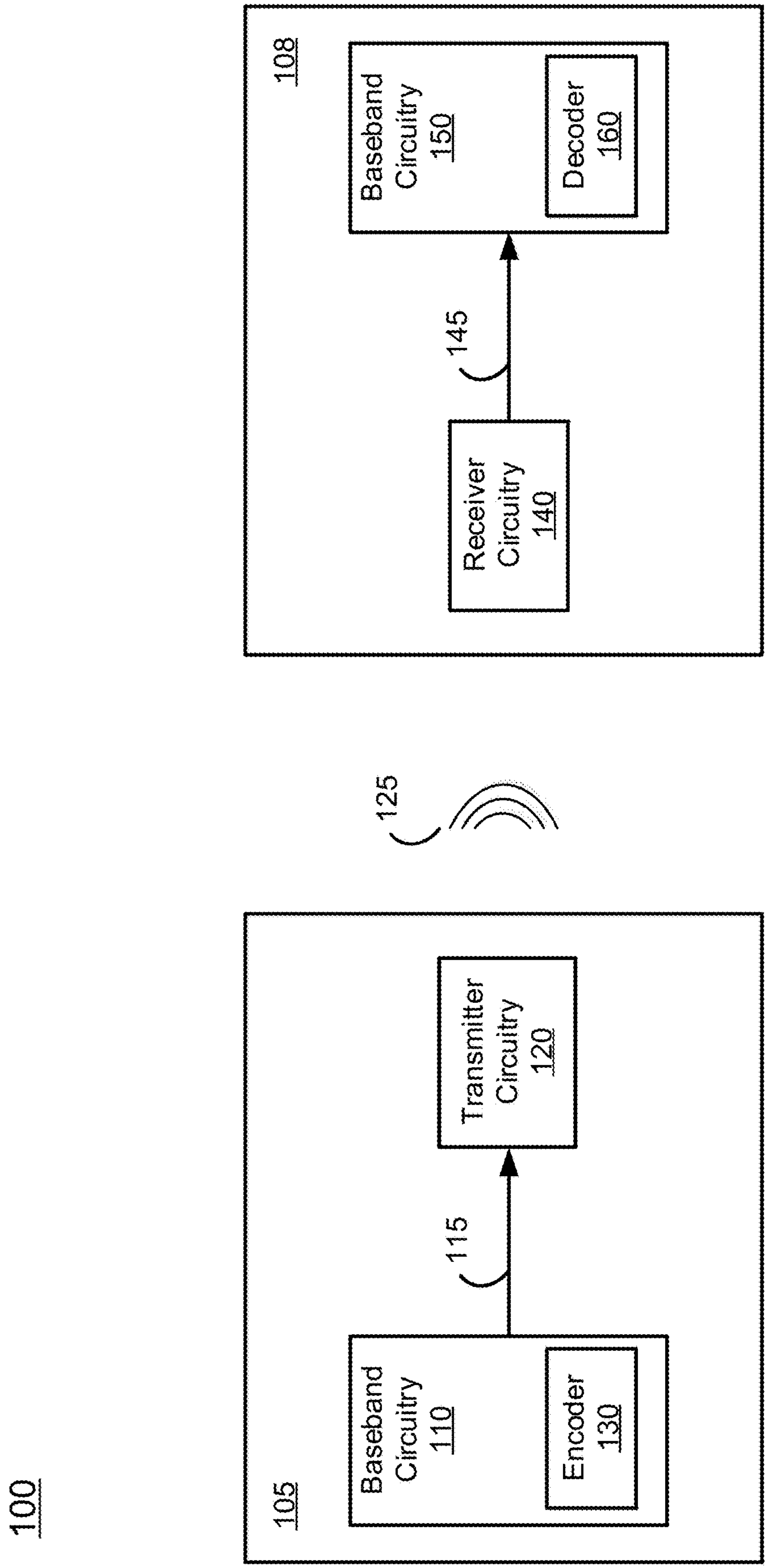

100
105
Baseband Circuitry 110
Encoder 130
115
Transmitter Circuitry 120
125
108
Receiver Circuitry 140
145
Baseband Circuitry 150
Decoder 160

FIG. 1

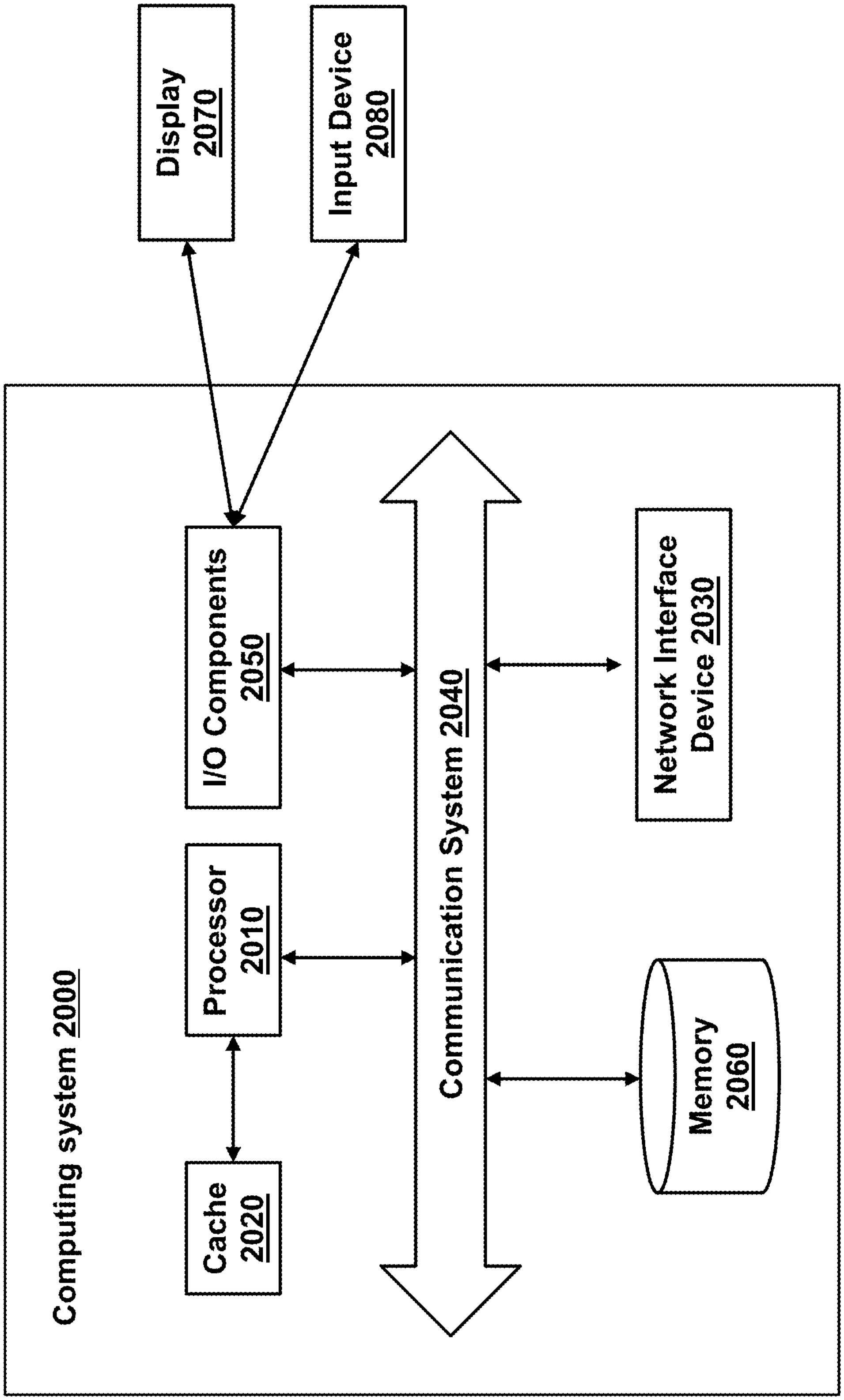

Display 2070
Input Device 2080
Computing system 2000
Cache 2020
Processor 2010
I/O Components 2050
Communication System 2040
Network Interface Device 2030
Memory 2060

FIG. 2

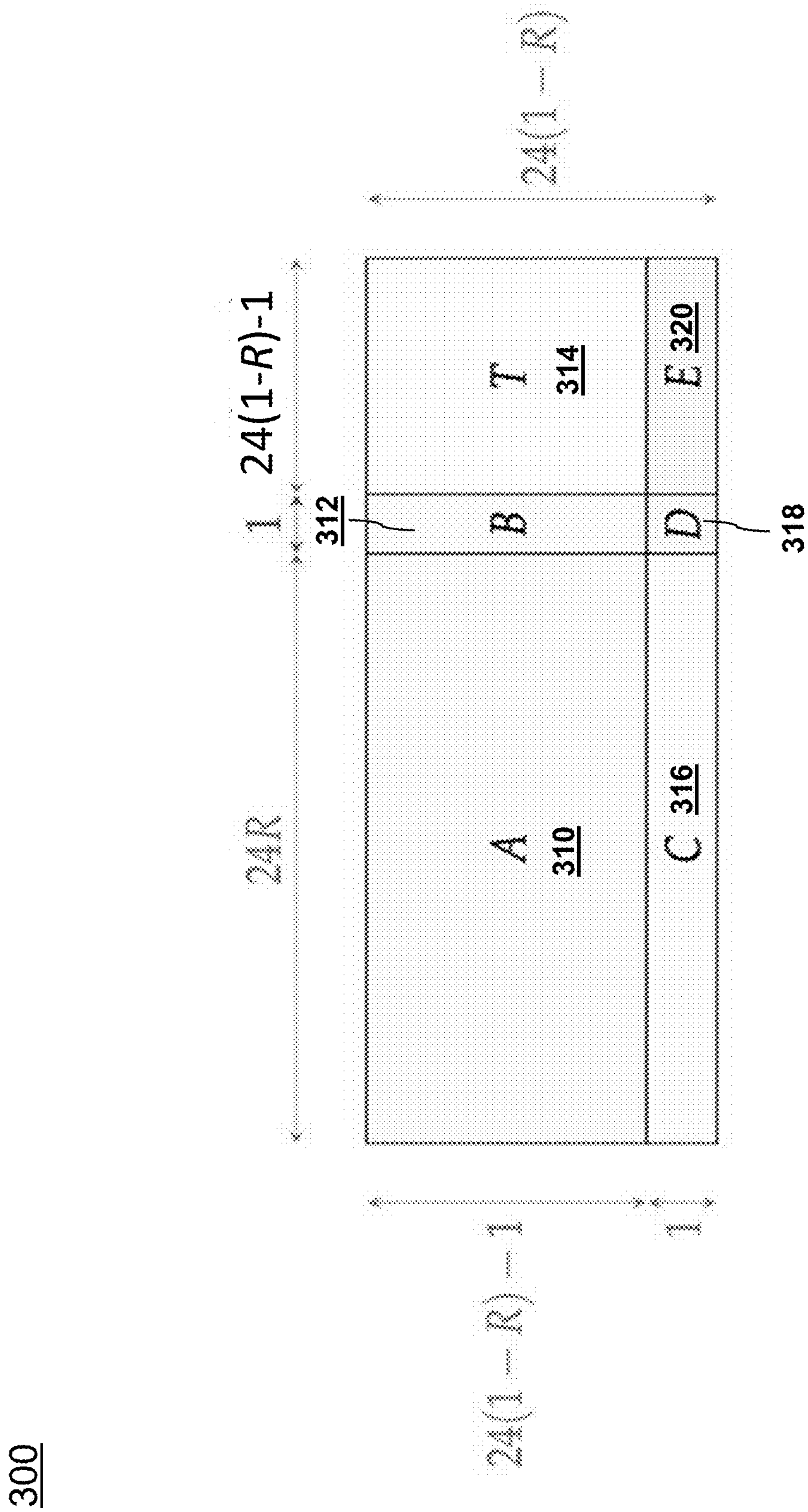


FIG. 3

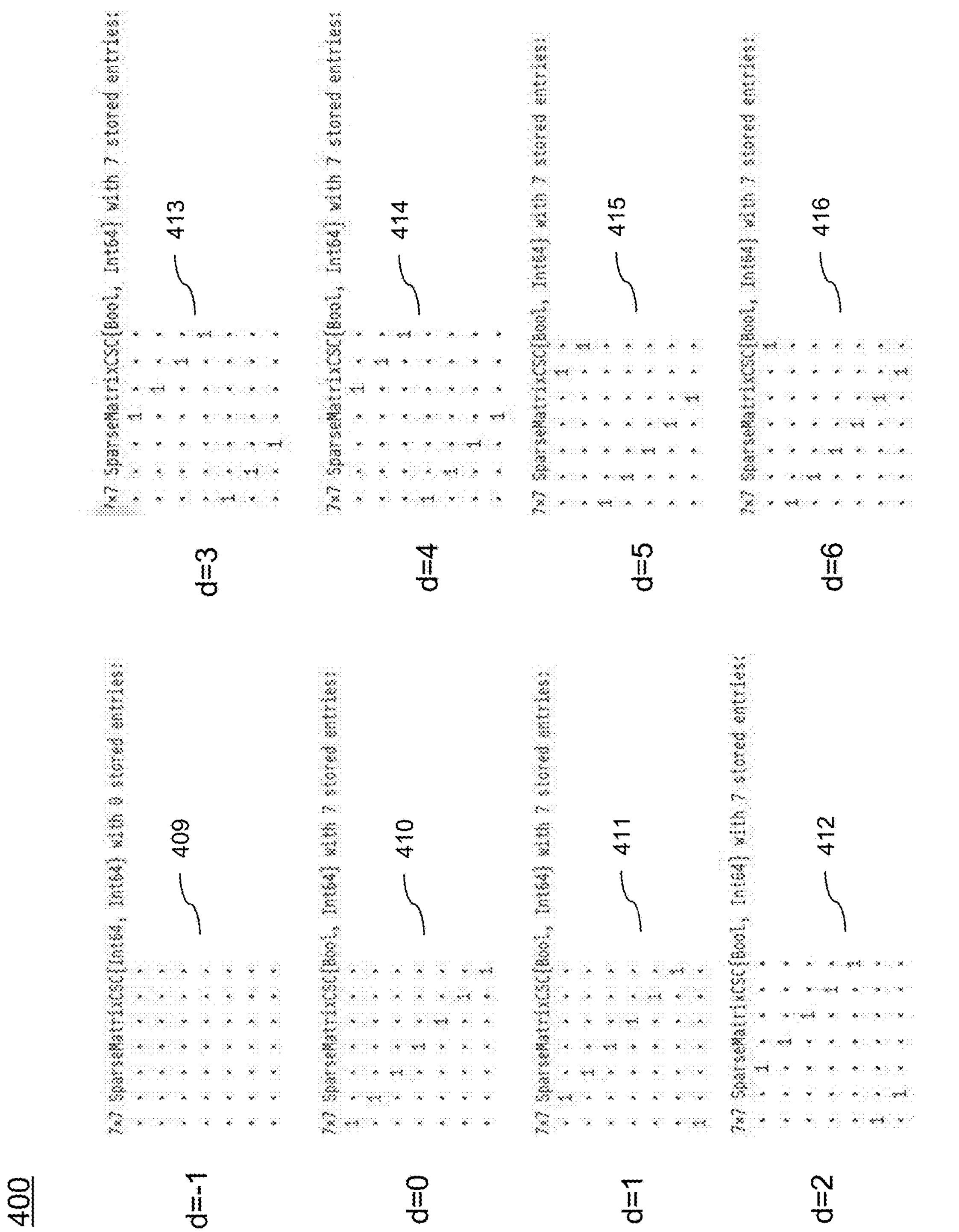

400
d=-1
7x7 SparseMatrixCSC{Int64, Int64} with 0 stored entries:
409
d=0
7x7 SparseMatrixCSC{Bool, Int64} with 7 stored entries:
410
d=1
7x7 SparseMatrixCSC{Bool, Int64} with 7 stored entries:
411
d=2
7x7 SparseMatrixCSC{Bool, Int64} with 7 stored entries:
412
d=3
7x7 SparseMatrixCSC{Bool, Int64} with 7 stored entries:
413
d=4
7x7 SparseMatrixCSC{Bool, Int64} with 7 stored entries:
414
d=5
7x7 SparseMatrixCSC{Bool, Int64} with 7 stored entries:
415
d=6
7x7 SparseMatrixCSC{Bool, Int64} with 7 stored entries:
416

FIG. 4

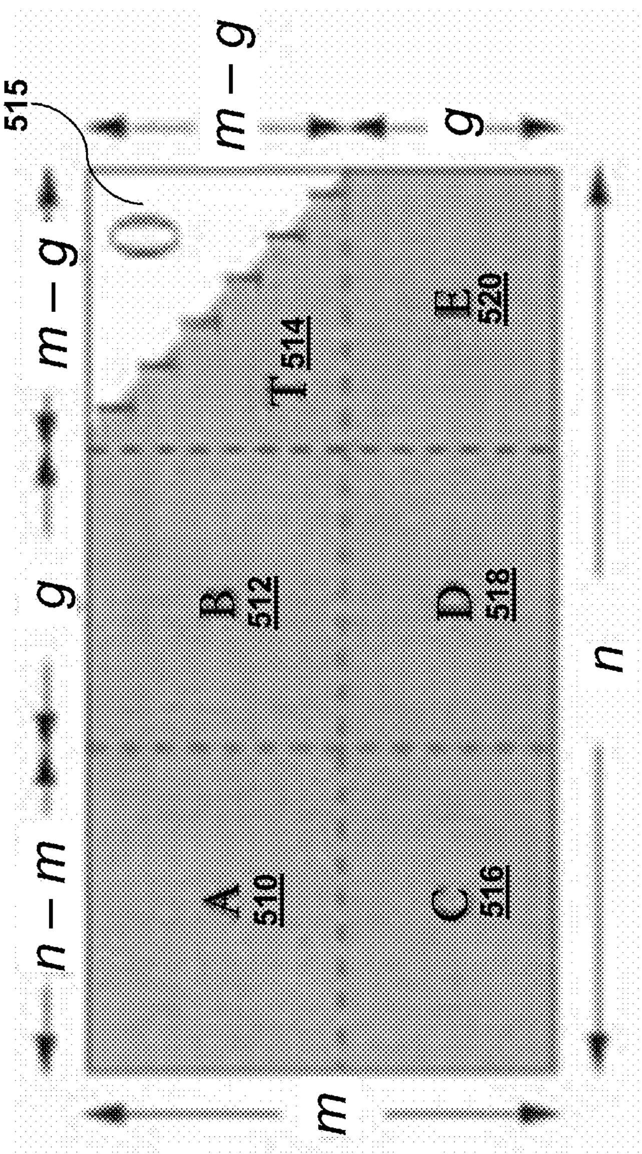


FIG. 5

500

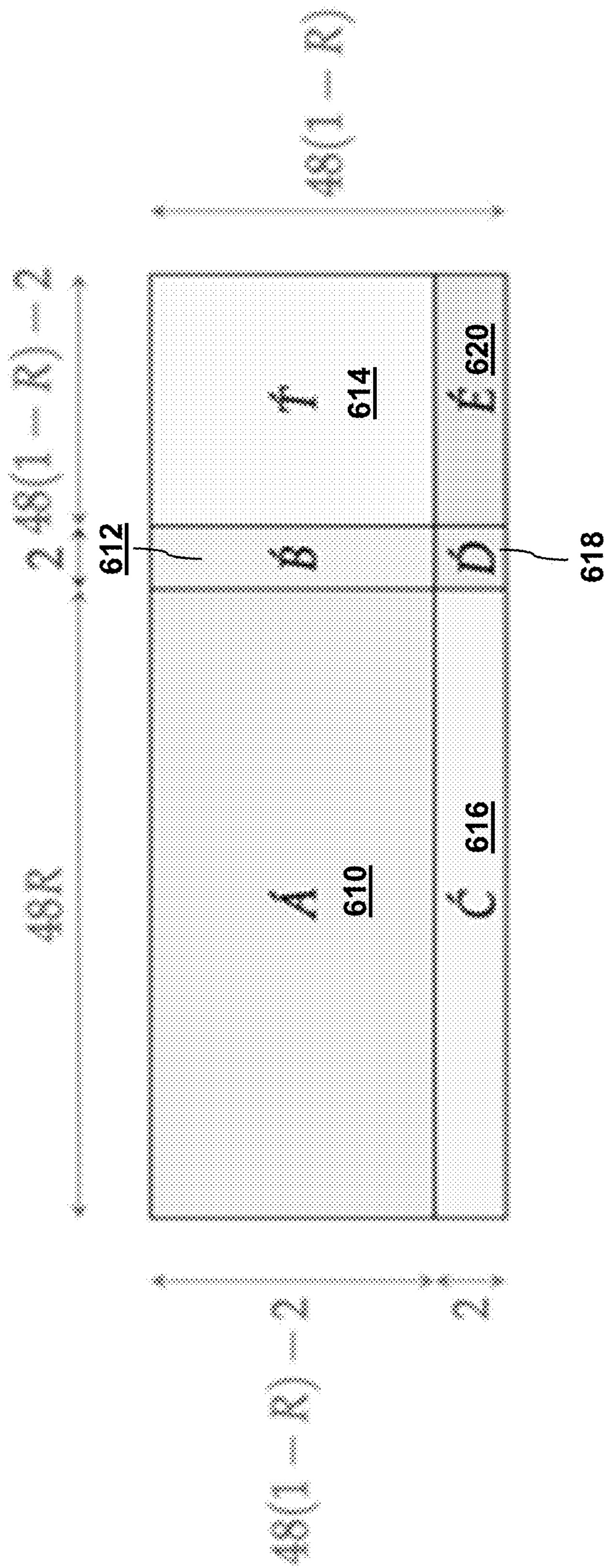


FIG. 6

600

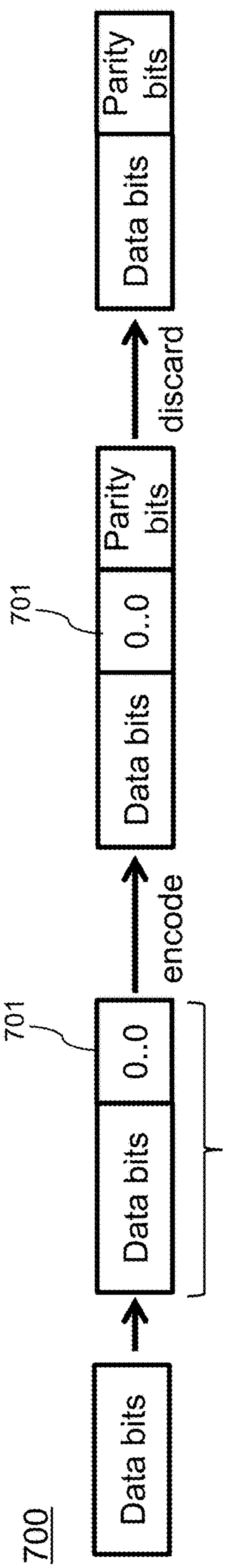


FIG. 7A

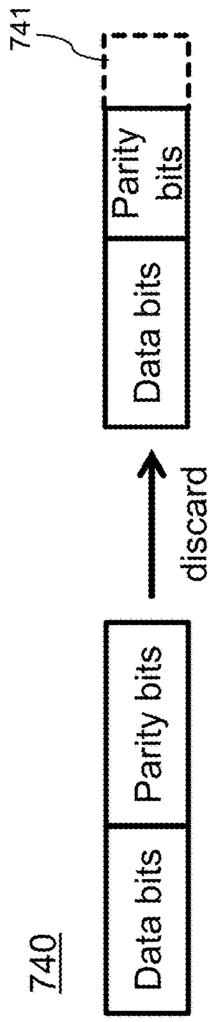


FIG. 7B

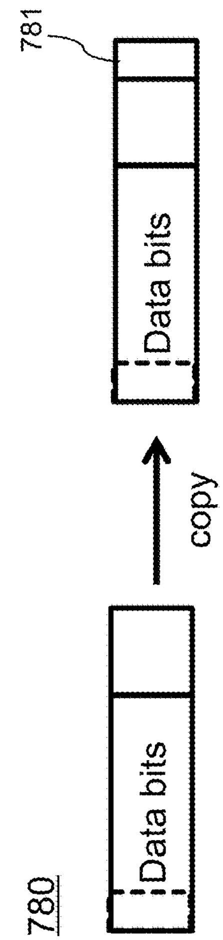
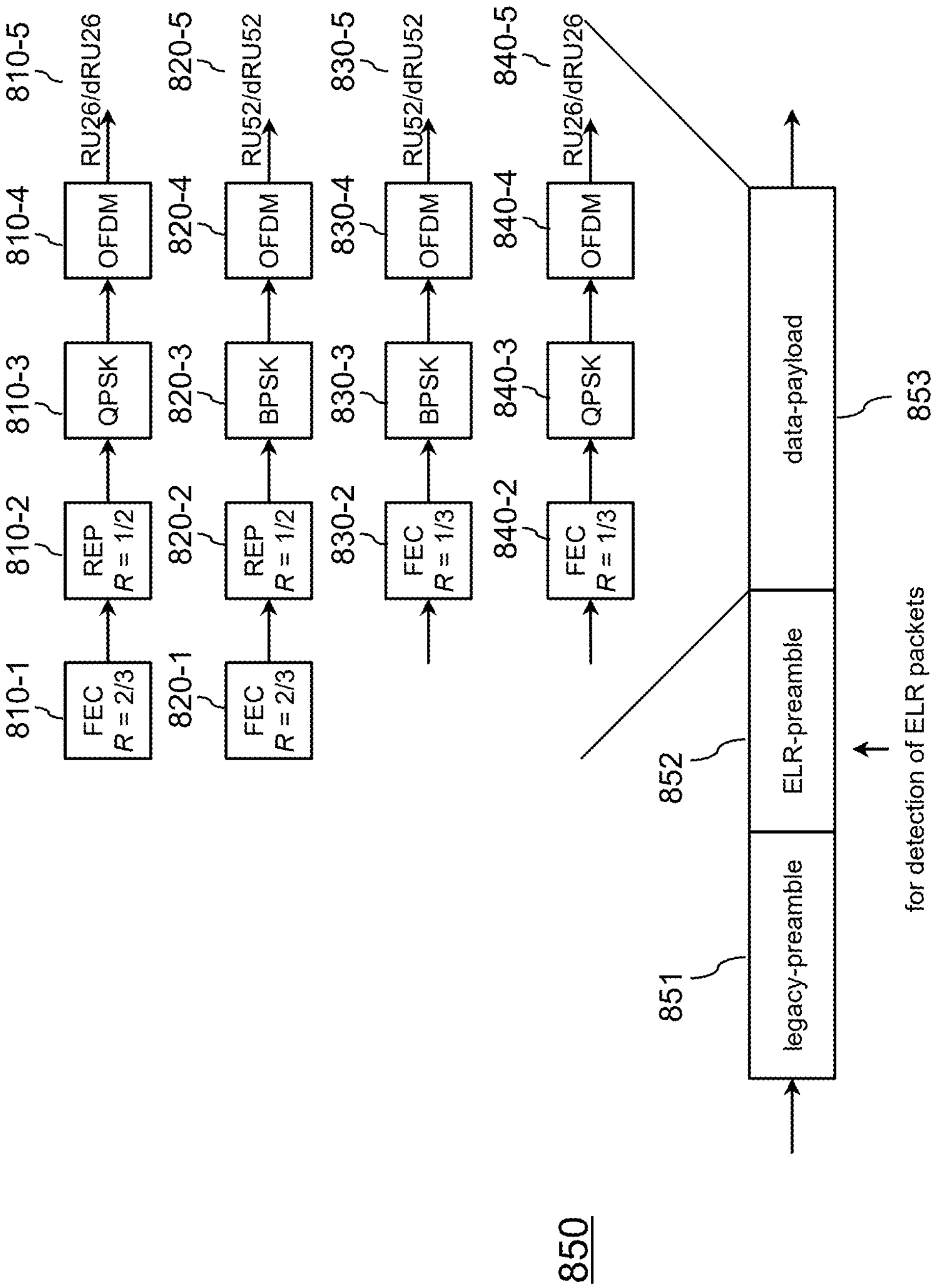
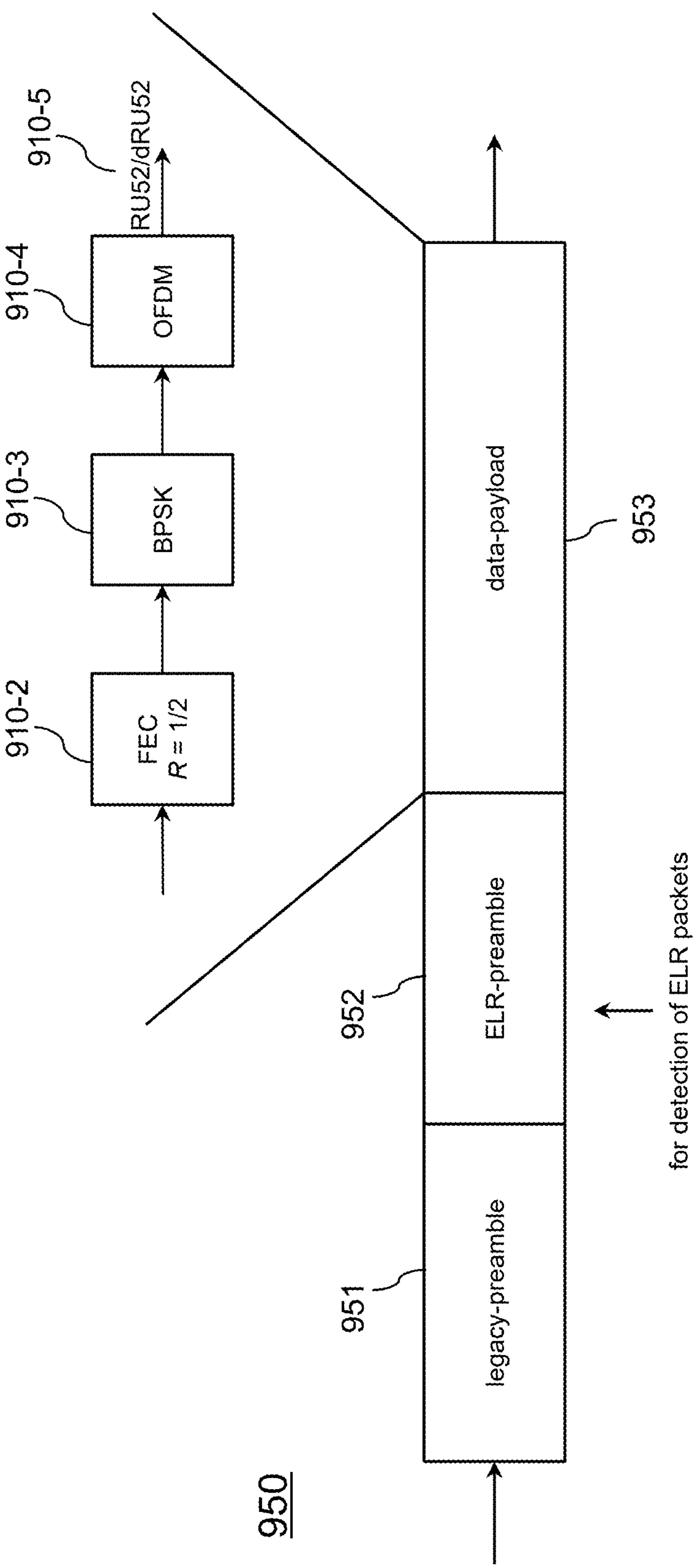


FIG. 7C



810-1
FEC
R = 2/3
810-2
REP
R = 1/2
810-3
QPSK
810-4
OFDM
810-5
RU26/dRU26
820-1
FEC
R = 2/3
820-2
REP
R = 1/2
820-3
BPSK
820-4
OFDM
820-5
RU52/dRU52
830-2
FEC
R = 1/3
830-3
BPSK
830-4
OFDM
830-5
RU52/dRU52
840-2
FEC
R = 1/3
840-3
QPSK
840-4
OFDM
840-5
RU26/dRU26
850
851
legacy-preamble
852
ELR-preamble
853
data-payload
for detection of ELR packets

FIG. 8



910-2
FEC
R = 1/2
910-3
BPSK
910-4
OFDM
910-5
RU52/dRU52
950
951
legacy-preamble
952
ELR-preamble
953
data-payload
for detection of ELR packets

FIG. 9

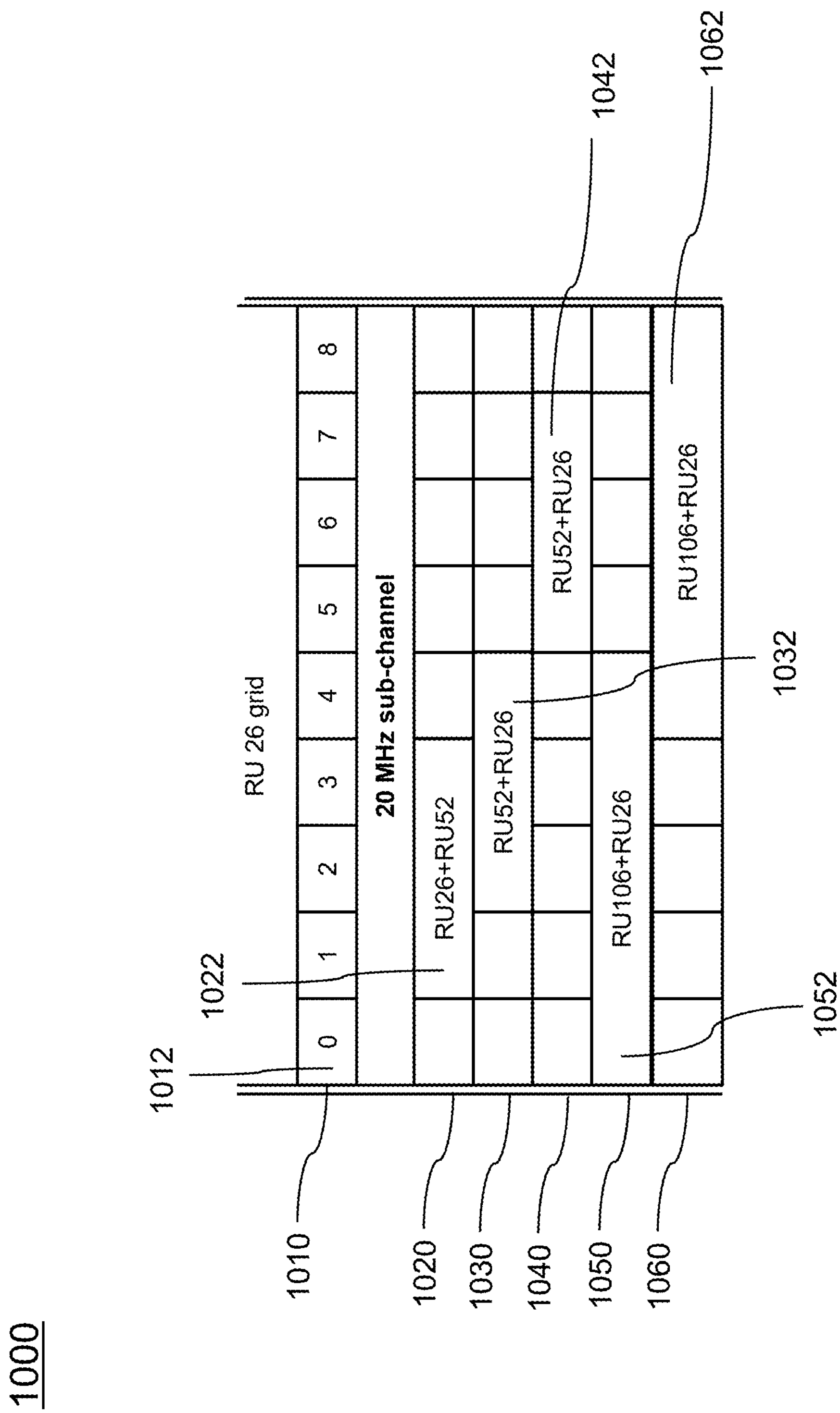

1000
RU 26 grid
0
1
2
3
4
5
6
7
8
20 MHz sub-channel
RU26+RU52
RU52+RU26
RU52+RU26
RU106+RU26
RU106+RU26
1010
1012
1020
1022
1030
1032
1040
1042
1050
1052
1060
1062

FIG. 10

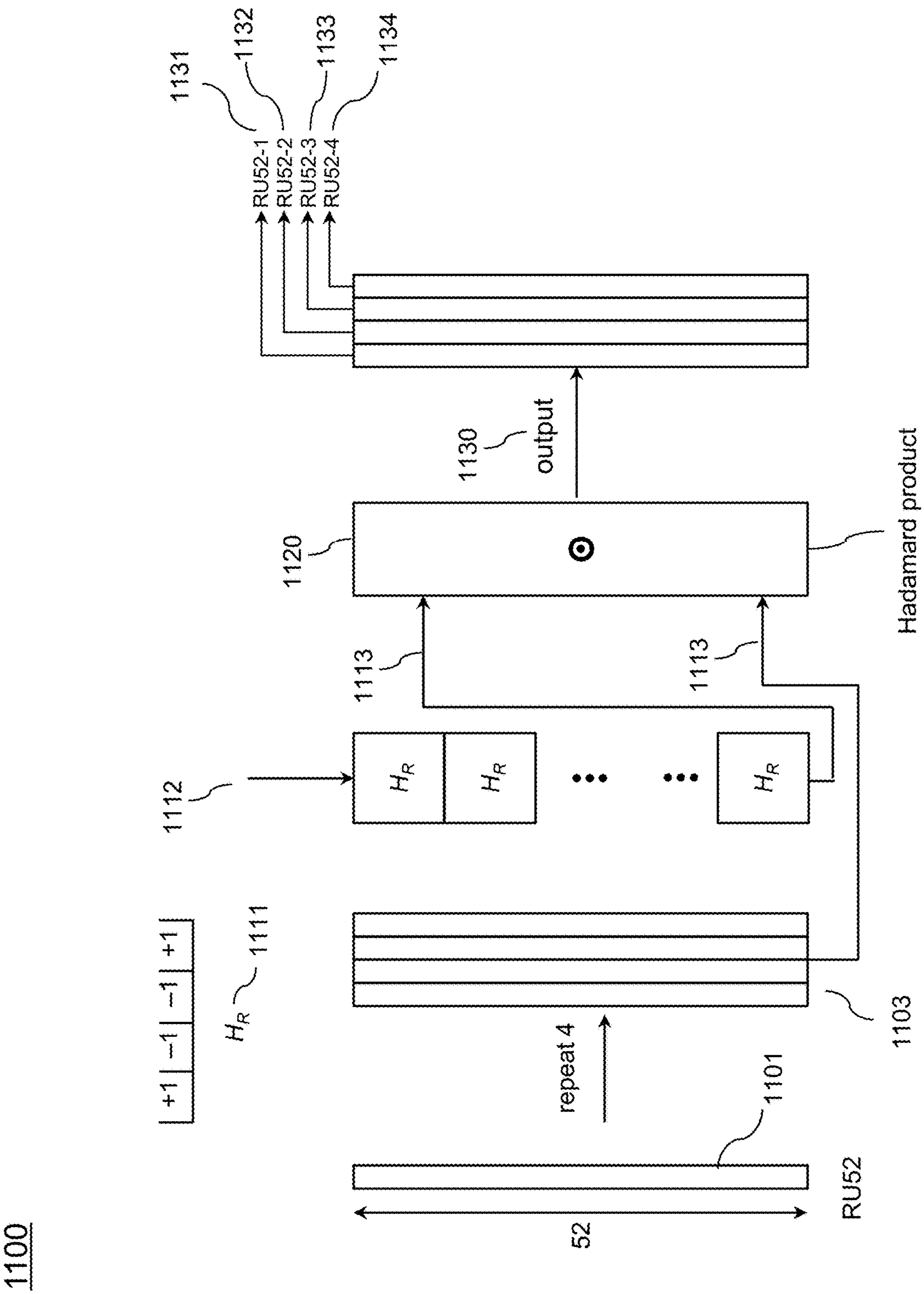

1100
52
RU52
1101
repeat 4
1103
H_R
1111
+1 | −1 | −1 | +1
1112
H_R
H_R
H_R
1113
1113
1120
⊙
Hadamard product
1130
output
RU52-1
RU52-2
RU52-3
RU52-4
1131
1132
1133
1134

FIG. 11

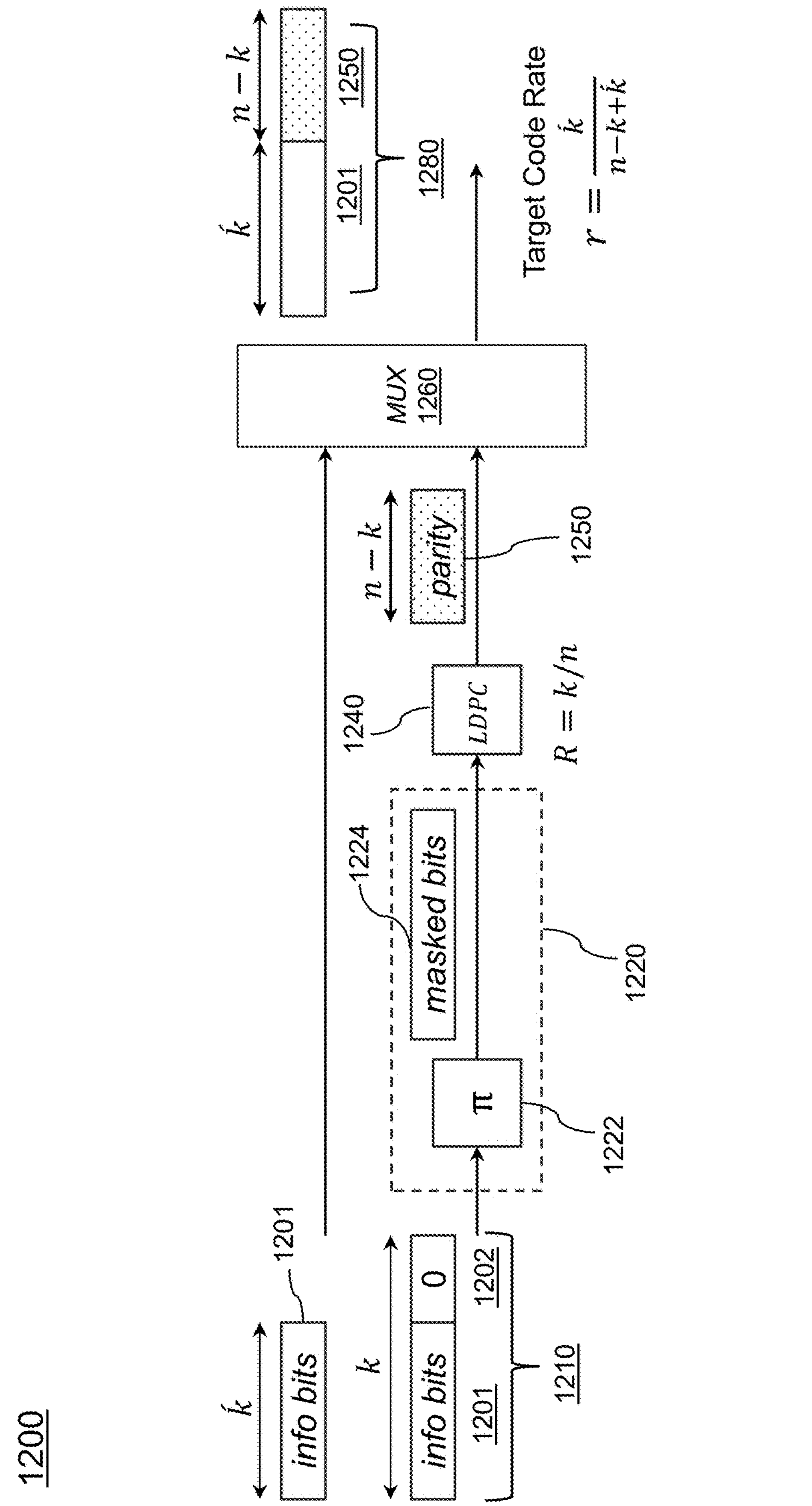
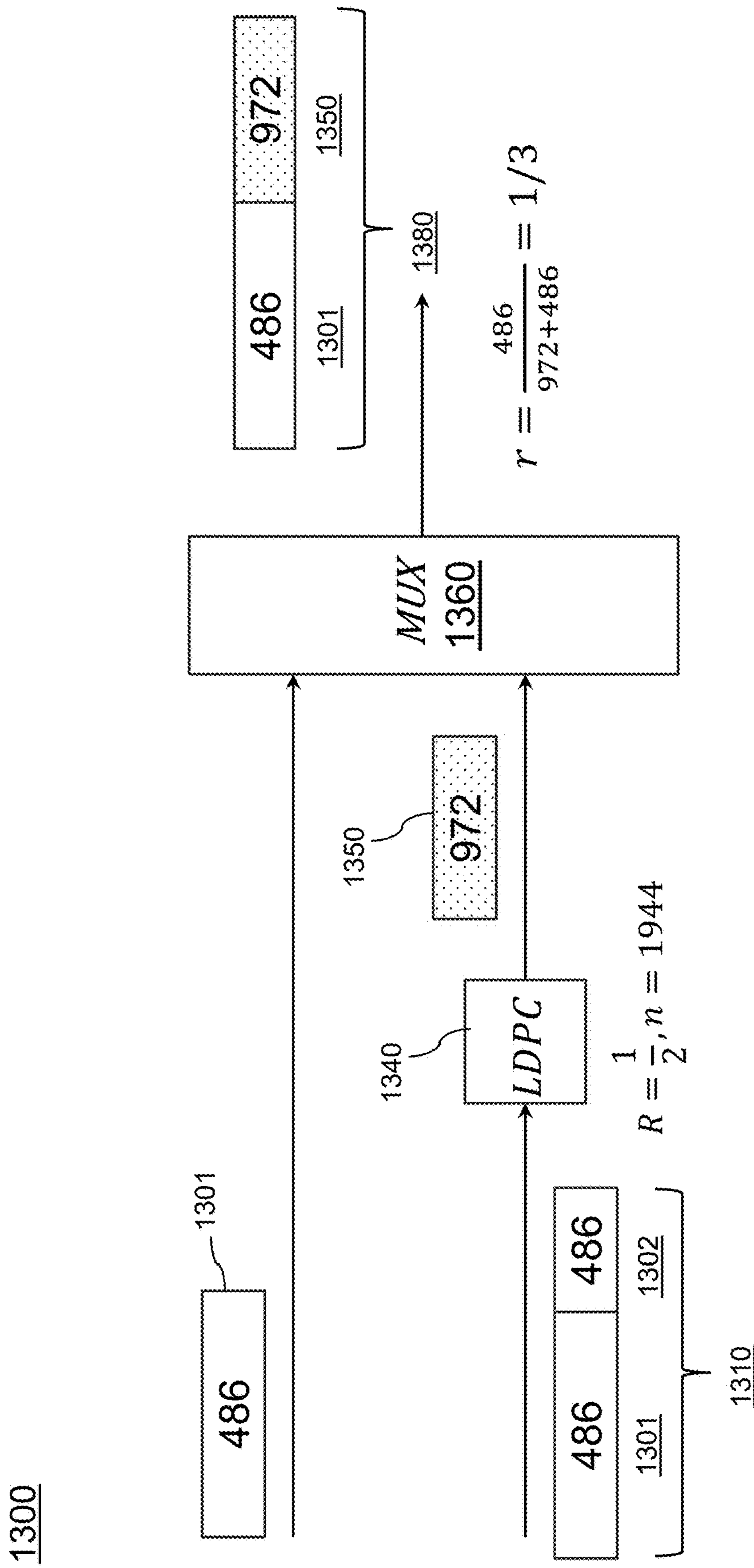

1200
info bits
1201
info bits
0
1201
1202
1210
π
1222
masked bits
1224
1220
LDPC
1240
R = k/n
parity
1250
MUX
1260
1201
1250
1280
Target Code Rate

FIG. 12



1300
486
1301
486
486
1301
1302
1310
LDPC
1340
$R = \frac{1}{2}, n = 1944$
972
1350
MUX
1360
486
972
1301
1350
1380
$r = \frac{486}{972+486} = 1/3$

FIG. 13

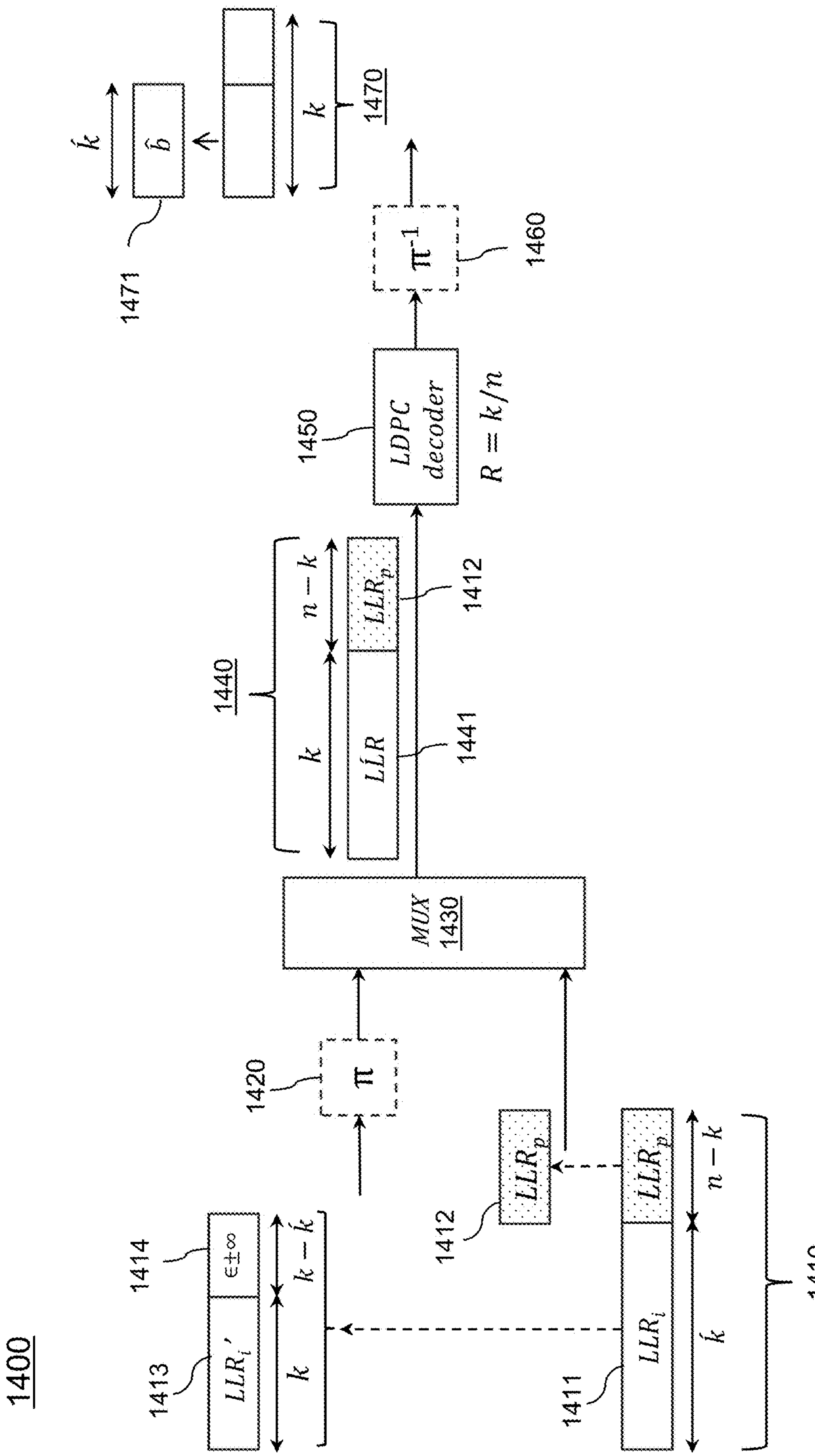

1400
1410
1411
1412
1413
1414
1420
1430
1440
1441
1450
1460
1470
1471
MUX
LDPC decoder
R = k/n
π
π-1
LLRi
LLRi'
LLRp
LLR
n − k
k

FIG. 14

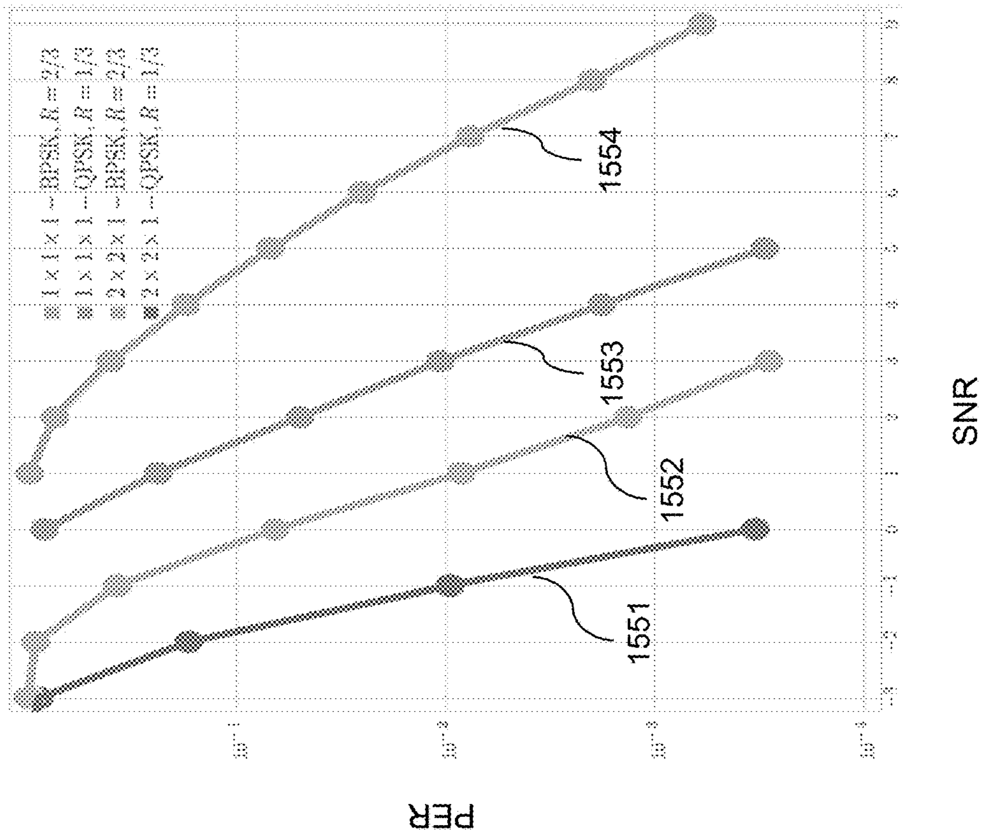


FIG. 15B

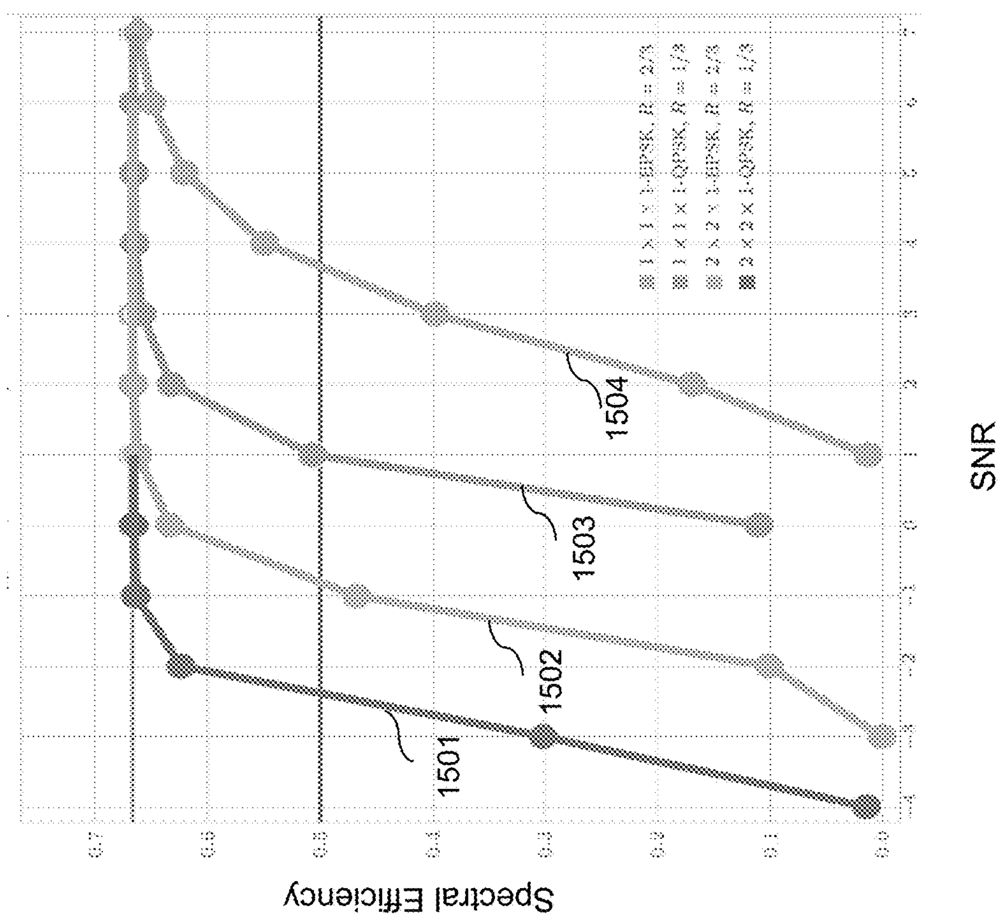


FIG. 15A

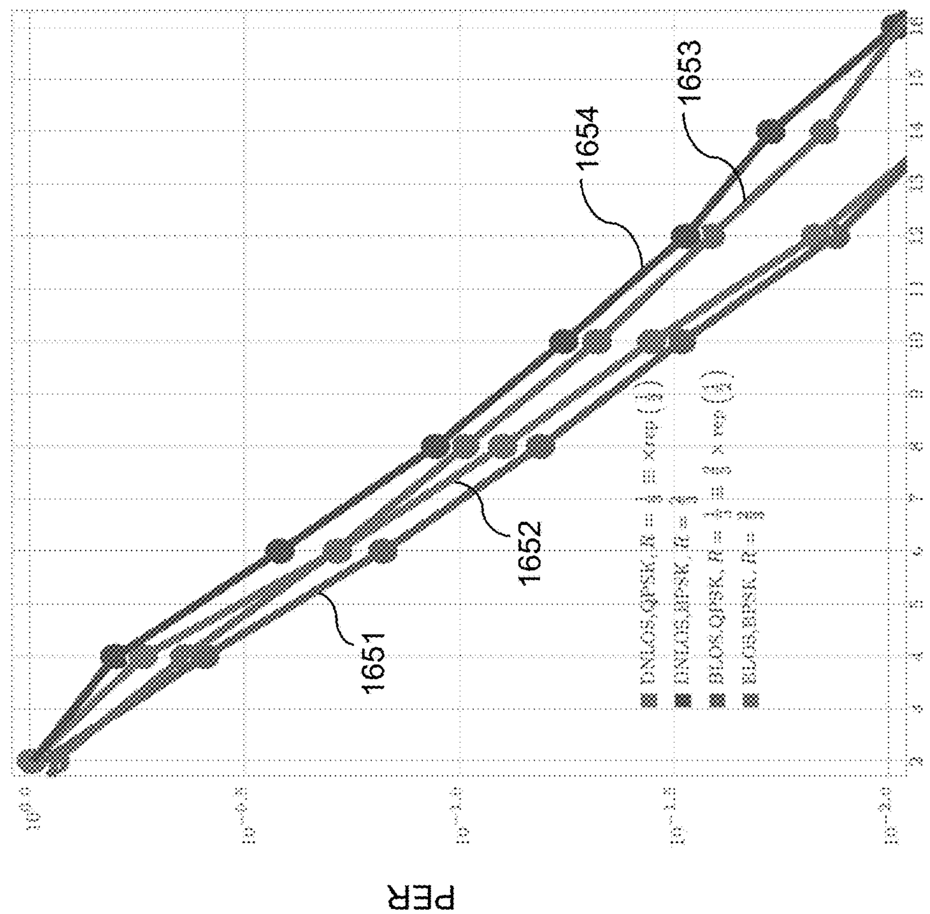


FIG. 16B

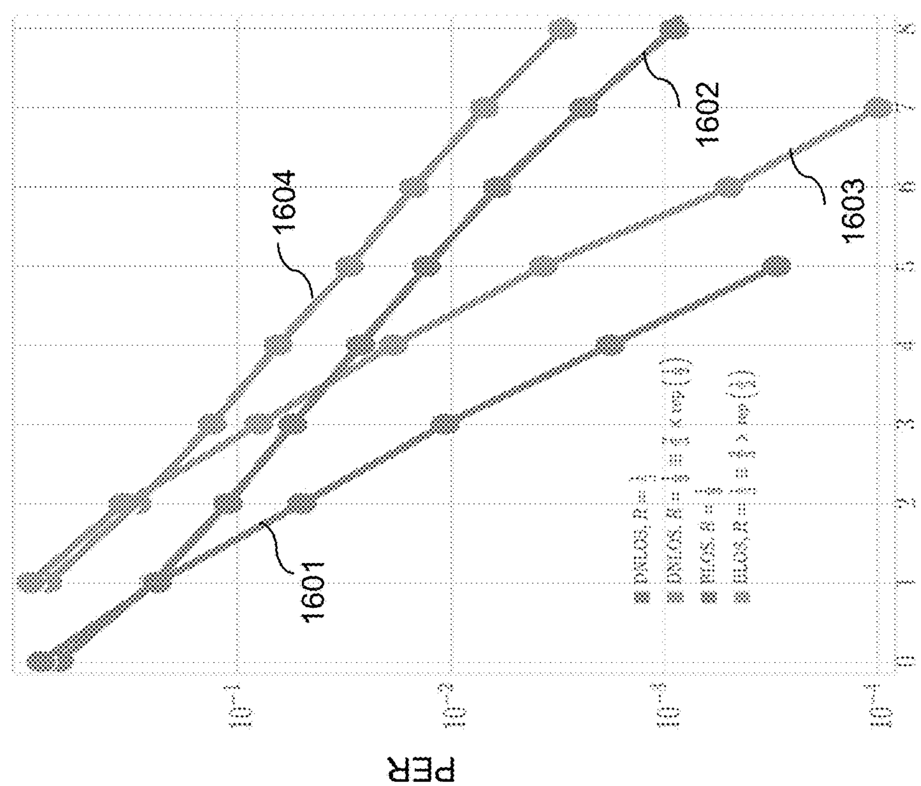


FIG. 16A

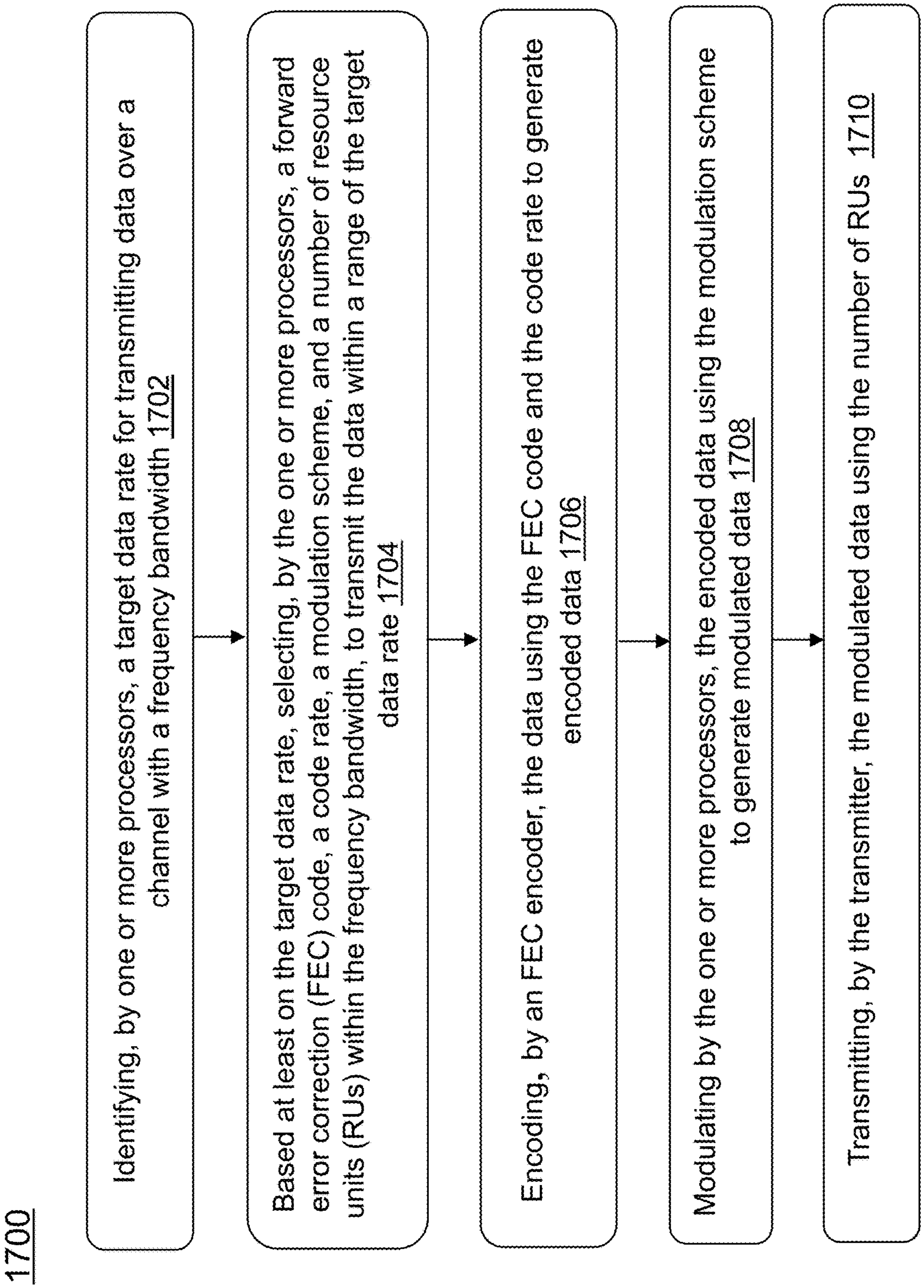

1700
Identifying, by one or more processors, a target data rate for transmitting data over a channel with a frequency bandwidth 1702
Based at least on the target data rate, selecting, by the one or more processors, a forward error correction (FEC) code, a code rate, a modulation scheme, and a number of resource units (RUs) within the frequency bandwidth, to transmit the data within a range of the target data rate 1704
Encoding, by an FEC encoder, the data using the FEC code and the code rate to generate encoded data 1706
Modulating by the one or more processors, the encoded data using the modulation scheme to generate modulated data 1708
Transmitting, by the transmitter, the modulated data using the number of RUs 1710

FIG. 17

SYSTEMS AND METHODS FOR EXTENDED LONG RANGE COMMUNICATION IN WIRELESS LOCAL AREA NETWORKS (WLANS)

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/560,191, filed on Mar. 1, 2024, the disclosure of which is incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] This disclosure generally relates to systems and methods for providing a mode to 2support PHY (physical layer) rate as low as 1 Mpbs, 1.1 Mbps, 1.2 Mbps, 1.5 Mbps, 1.7 Mbps or their multiples in a wireless local area network (WLAN) to achieve extended (or enhanced) long range (ELR) communication.

BACKGROUND OF THE DISCLOSURE

[0003] The market for wireless communications devices has been growing due to increased use of portable devices, increased connectivity and data transfer between all manners of devices. Digital switching techniques have facilitated the large scale deployment of affordable, easy-to-use wireless communication networks. Wireless communication can operate in accordance with various standards, such as the IEEE 802.11x (e.g., Wi-Fi technology), Bluetooth, global system for mobile communications (GSM), code division multiple access (CDMA). Using such technologies, wireless communication devices can connect to local area networks and the internet without physical cables, communicating over radio frequencies and across various spaces and ranges.

[0004] Ultra High Reliability (UHR) is a new study group within the IEEE 802.11 working group that focuses on enhancing the reliability of Wireless Local Area Network (WLAN) connectivity. Extended (or Enhanced) Long Range (ELR) refers to communication schemes designed to extend the range and reliability of WLANs. ERL communication schemes can be implemented based on frequency domain duplication or repeating the tones with or without rotation. An orthogonal frequency division multiplexing (OFDM) channel may include a plurality of subcarriers (tones). These tones can be grouped into smaller sub-channels called Resource Units (RUs). A code rate (or coding rate) represents the ratio of useful information (data) bits to the total transmitted bits. A code rate can have a great impact on the efficiency of data transmission. For example, a modulation coding scheme (MCS) over RU26 (resource units each containing 24 data tones and 2 pilot tones, for example) can support data rates 800-900 Kbps depending on cyclic prefix (CP) size. Low-density parity check (LDPC) codes can be used for designs/implementations in communication standards (e.g., IEEE 802.11bn UHR chips).

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

[0006] FIG. **1** is a diagram depicting an example communication environment with communication systems, according to one or more embodiments.

[0007] FIG. **2** is a schematic block diagram of a computing system, according to an embodiment.

[0008] FIG. **3** is a diagram depicting an example exponent matrix, according to one or more embodiments.

[0009] FIG. **4** is a diagram depicting example shifted identity matrices for generating a parity check matrix, according to one or more embodiments.

[0010] FIG. **5** is a diagram depicting an example parity check matrix, according to one or more embodiments.

[0011] FIG. **6** is a diagram depicting an example exponent matrix, according to one or more embodiments.

[0012] FIG. **7**A, FIG. **7**B, and FIG. **7**C are diagrams depicting example rate matching schemes including shortening, puncturing, and repeating, respectively, according to one or more embodiments.

[0013] FIG. **8** is a diagram depicting a structure of an ELR packet and various examples of schemes for encoding, modulating and transmitting the ELR packet to obtain a target data rate, according to one or more implementations.

[0014] FIG. **9** is a diagram depicting a structure of an ELR packet and another example of schemes for encoding, modulating and transmitting the ELR packet to obtain a target data rate, according to one or more implementations.

[0015] FIG. **10** is a diagram depicting an allocation structure of resource units (RUs) in a bandwidth of 20 MHz, according to one or more implementations.

[0016] FIG. **11** is a diagram depicting a structure of repeating RUs in a bandwidth of 20 MHz, according to one or more implementations.

[0017] FIG. **12** is a diagram depicting an example encoder for encoding with a shortened LDPC code to obtain a particular code rate, according to one or more embodiments.

[0018] FIG. **13** is a diagram depicting an example encoder for encoding with a shortened LDPC code to obtain a particular code rate, according to one or more embodiments

[0019] FIG. **14** is a diagram depicting an example decoder for decoding with a shortened LDPC code according to one or more embodiments.

[0020] FIG. **15**A and FIG. **15**B are diagrams depicting example simulation results using a system to obtain a target data rate, according to one or more embodiments.

[0021] FIG. **16**A and FIG. **16**B are diagrams depicting example simulation results using a system to obtain a target data rate, according to one or more embodiments.

[0022] FIG. **17** is a flow diagram showing a process for obtaining a particular data rate using various schemes of encoding, modulation and/or transmitting, in accordance with an embodiment.

[0023] The details of various embodiments of the methods and systems are set forth in the accompanying drawings and the description below.

DETAILED DESCRIPTION

[0024] The following IEEE standard(s), including any draft versions of such standard(s), are hereby incorporated herein by reference in their entirety and are made part of the present disclosure for all purposes: Wi-Fi Alliance standards and IEEE 802.11 standards including but not limited to

IEEE 802.11a™, IEEE 802.11b™, IEEE 802.11g™, IEEE P802.11n™; IEEE P802.11ac™; and IEEE P802.11be™ through IEEE P802.11bn™ standards. Although this disclosure can reference aspects of these standard(s), the disclosure is in no way limited by these standard(s).

[0025] For purposes of reading the description of the various embodiments below, the following descriptions of the sections of the specification and their respective contents can be helpful:

- **[0026]** Section A describes a network environment and computing environment which can be useful for practicing embodiments described herein;
- **[0027]** Section B describes LDPC-based encoding/decoding systems which can be useful for practicing embodiments described herein; and
- **[0028]** Section C describes embodiments of systems and methods to achieve extended long range communication to provide an aggregate throughput as low as 1 Mbps in a WLAN.

A. Computing and Network Environment

[0029] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, a first feature in communication with or communicatively coupled to a second feature in the description that follows may include embodiments in which the first feature is in direct communication with or directly coupled to the second feature and may also include embodiments in which additional features may intervene between the first and second features, such that the first feature is in indirect communication with or indirectly coupled to the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0030] Various embodiments disclosed herein are related to one or more apparatuses, devices and/or systems, including a transmitter and/or receiver and one or more processors and that may be configured, constructed or implemented to communicate using any encoding process and techniques as defined or supported by any IEEE 801.11 standard such as 902.11n, 802.11AC, 802.11ax and 802.11be or other versions and embodiments of the IEEE 802.11 standard.

[0031] Referring to FIG. **1**, illustrated is a diagram depicting an example communication environment **100** including communication systems (or communication apparatuses) **105**, **108**, according to one or more embodiments. In one embodiment, the communication system **105** includes a baseband circuitry **110** and a transmitter circuitry **120**, and the communication system **108** includes a baseband circuitry **150** and a receiver circuitry **140**. In one aspect, the communication system **105** is considered a transmitter communication system, and the communication system **108** is considered a receiver communication system. These components operate together to exchange data (e.g., messages or frames) through a wireless medium. These components are embodied as application specific integrated circuit (ASIC), field programmable gate array (FPGA), or any combination of these, in one or more embodiments. In some embodiments, the communication systems **105**, **108** include more, fewer, or different components than shown in FIG. **1**. For example, each of the communication systems **105**, **108** includes transceiver circuitry to allow bi-directional communication between the communication systems **105**, **108** or with other communication systems. In some embodiments, each of the communication systems **105**, **108** may have configuration similar to that of a computing system **2000** as shown in FIG. **2**.

[0032] The baseband circuitry **110** of the communication system **105** is a circuitry that generates the baseband data **115** for transmission. The baseband data **115** includes information data (e.g., signal(s)) at a baseband frequency for transmission. In one approach, the baseband circuitry **110** includes an encoder **130** that encodes the data, and generates or outputs parity bits. In one aspect, the baseband circuitry **110** (or encoder **130**) obtains a generator matrix or a parity check matrix, or uses a previously produced generator matrix or a previously produced parity check matrix, and encodes the information data by applying the information data to the generator matrix or the parity check matrix to obtain a codeword. In some embodiments, the baseband circuitry **110** stores one or more generator matrices or one or more parity check matrices that conform to any IEEE 802.11 standard for WLAN communication. The baseband circuitry **110** retrieves the stored generator matrix or the stored parity check matrix in response to detecting information data to be transmitted, or in response to receiving an instruction to encode the information data. In one approach, the baseband circuitry **110** generates the parity bits according to a portion of the generator matrix or using the parity check matrix, and appends the parity bits to the information bits to form a codeword. The baseband circuitry **110** generates the baseband data **115** including the codeword for the communication system **108**, and provides the baseband data **115** to the transmitter circuitry **120**.

[0033] The transmitter circuitry **120** of the communication system **105** includes or corresponds to a circuitry that receives the baseband data **115** from the baseband circuitry **110** and transmits a wireless signal **125** according to the baseband data **115**. In one configuration, the transmitter circuitry **120** is coupled between the baseband circuitry **110** and an antenna (not shown). In this configuration, the transmitter circuitry **120** up-converts the baseband data **115** from the baseband circuitry **110** onto a carrier signal to generate the wireless signal **125** at an RF frequency (e.g., 10 MHz to 60 GHz), and transmits the wireless signal **125** through the antenna.

[0034] The receiver circuitry **140** of the communication system **108** is a circuitry that receives the wireless signal **125** from the communication system **105** and obtains baseband data **145** from the received wireless signal **125**. In one configuration, the receiver circuitry **140** is coupled between the baseband circuitry **150** and an antenna (not shown). In this configuration, the receiver circuitry **140** receives the wireless signal **125** though an antenna, and down-converts the wireless signal **125** at an RF frequency according to a carrier signal to obtain the baseband data **145** from the wireless signal **125**. The receiver circuitry **140** then provides the baseband data **145** to the baseband circuitry **150**.

[0035] The baseband circuitry **150** of the communication system **108** includes or corresponds to a circuitry that receives the baseband data **145** from the receiver circuitry **140** and obtains information data from the received baseband data **145**. In one embodiment, the baseband circuitry **150** includes a decoder **160** that extracts information and parity bits from the baseband data **145**. The decoder **160** decodes the baseband data **145** to obtain the information data generated by the baseband circuitry **110** of the communication system **105**.

[0036] In some embodiments, each of the baseband circuitry **110** (including the encoder **130**), the transmitter circuitry **120**, the receiver circuitry **140**, and the baseband circuitry **150** (including the decoder **160**) may be as one or more processors, application specific integrated circuit (ASIC), field programmable gate array (FPGA), or any combination of them.

[0037] FIG. **2** is a schematic block diagram of a computing system, according to an embodiment. An illustrated example computing system **2000** includes one or more processors **2010** in direct or indirect communication, via a communication system **2040** (e.g., bus), with memory **2060**, at least one network interface controller **2030** with network interface port for connection to a network (not shown), and other components, e.g., input/output ("I/O") components **2050**. Generally, the processor(s) **2010** will execute instructions (or computer programs) received from memory. The processor(s) **2010** illustrated incorporate, or are connected to, cache memory **2020**. In some instances, instructions are read from memory **2060** into cache memory **2020** and executed by the processor(s) **2010** from cache memory **2020**. The computing system **2000** may not necessarily contain all of these components shown in FIG. **2**, and may contain other components that are not shown in FIG. **2**.

[0038] In more detail, the processor(s) **2010** may be any logic circuitry that processes instructions, e.g., instructions fetched from the memory **2060** or cache **2020**. In many implementations, the processor(s) **2010** are microprocessor units or special purpose processors. The computing device **2050** may be based on any processor, or set of processors, capable of operating as described herein. The processor(s) **2010** may be single core or multi-core processor(s). The processor(s) **2010** may be multiple distinct processors.

[0039] The memory **2060** may be any device suitable for storing computer readable data. The memory **2060** may be a device with fixed storage or a device for reading removable storage media. Examples include all forms of volatile memory (e.g., RAM), non-volatile memory, media and memory devices, semiconductor memory devices (e.g., EPROM, EEPROM, SDRAM, and flash memory devices), magnetic disks, magneto optical disks, and optical discs (e.g., CD ROM, DVD-ROM, or Blu-Ray® discs). A computing system **2000** may have any number of memory devices **2060**.

[0040] The cache memory **2020** is generally a form of computer memory placed in close proximity to the processor (s) **2010** for fast read times. In some implementations, the cache memory **2020** is part of, or on the same chip as, the processor(s) **2010**. In some implementations, there are multiple levels of cache **2020**, e.g., L2 and L3 cache layers.

[0041] The network interface controller **2030** manages data exchanges via the network interface (sometimes referred to as network interface ports). The network interface controller **2030** handles the physical and data link layers of the OSI model for network communication. In some implementations, some of the network interface controller's tasks are handled by one or more of the processor(s) **2010**. In some implementations, the network interface controller **2030** is part of a processor **2010**. In some implementations, the computing system **2000** has multiple network interfaces controlled by a single controller **2030**. In some implementations, the computing system **2000** has multiple network interface controllers **2030**. In some implementations, each network interface is a connection point for a physical network link (e.g., a cat-5 Ethernet link). In some implementations, the network interface controller **2030** supports wireless network connections and an interface port is a wireless (e.g., radio) receiver or transmitter (e.g., for any of the IEEE 802.11 protocols, near field communication "NFC", Bluetooth, ANT, or any other wireless protocol). In some implementations, the network interface controller **2030** implements one or more network protocols such as Ethernet. Generally, a computing device **2050** exchanges data with other computing devices via physical or wireless links through a network interface. The network interface may link directly to another device or to another device via an intermediary device, e.g., a network device such as a hub, a bridge, a switch, or a router, connecting the computing device **2000** to a data network such as the Internet.

[0042] The computing system **2000** may include, or provide interfaces for, one or more input or output ("I/O") devices. Input devices include, without limitation, keyboards, microphones, touch screens, foot pedals, sensors, MIDI devices, and pointing devices such as a mouse or trackball. Output devices include, without limitation, video displays, speakers, refreshable Braille terminal, lights, MIDI devices, and 2-D or 3-D printers.

[0043] Other components may include an I/O interface, external serial device ports, and any additional co-processors. For example, a computing system **2000** may include an interface (e.g., a universal serial bus (USB) interface) for connecting input devices, output devices, or additional memory devices (e.g., portable flash drive or external media drive). In some implementations, a computing device **2000** includes an additional device such as a co-processor, e.g., a math co-processor can assist the processor **2010** with high precision or complex calculations.

[0044] The components **2090** may be configured to connect with external media, a display **2070**, an input device **2080** or any other components in the computing system **2000**, or combinations thereof. The display **2070** may be a liquid crystal display (LCD), an organic light emitting diode (OLED) display, a flat panel display, a solid state display, a cathode ray tube (CRT) display, a projector, a printer or other now known or later developed display device for outputting determined information. The display **2070** may act as an interface for the user to see the functioning of the processor (s) **2010**, or specifically as an interface with the software stored in the memory **2060**.

[0045] The input device **2080** may be configured to allow a user to interact with any of the components of the computing system **2000**. The input device **2080** may be a plurality pad, a keyboard, a cursor control device, such as a mouse, or a joystick. Also, the input device **2080** may be a remote control, touchscreen display (which may be a combination of the display **2070** and the input device **2080**), or any other device operative to interact with the computing system **2000**, such as any device operative to act as an interface between a user and the computing system **2000**.

B. LDPC-Based Encoding/Decoding Systems

[0046] Generally, a parity check matrix for a code represents equations that determine whether errors have occurred during transmission. More formally, for all valid codewords (i.e., bits produced by the encoder with no errors), the following equation can be true:

$$Hc = 0 \qquad \text{(Equation 1)}$$

[0047] In Equation 1, "H" is the parity check matrix, "c" is a codeword vector, and "0" is a vector of all zeroes. The parity check matrix, H, is one way of describing a code.

[0048] A generator matrix for a code, G, satisfies the following equation:

$$sG = c \qquad \text{(Equation 2)}$$

[0049] In Equation 2, "s" is a vector of information bits, "G" is a generator matrix and "c" is the codeword that corresponds to "s." In some implementations, a system (e.g., a communication system **108** including a decoder **160** in FIG. **1**) can decode the codeword c to obtain the decoded data s using Equation 2.

[0050] The parity check and generator matrices for a code are related per the above matrix equations. Generally, if a parity check matrix is low density, the corresponding generator matrix will be high density, and vice versa. LCPC codes are accordingly characterized by low density parity check matrices and high density generator matrices. The density of a matrix relates to the number of operations that must be performed to implement one of the above equations. Although it was recognized by 1995 that LDPC codes could be used to transmit data with very few errors, i.e., with error rates as good or better than turbo codes, one disadvantage of LDPC codes is that their generator matrices were high density and that made encoding computationally intensive, rending the codes impractical for many applications.

[0051] In some implementations, a parity check matrix may have a quasi-cyclic structure, for example, a parity check matrix for QC-LDPC code (n=3888, k=2916, R=3/4). Given a lifting size z, the parity check matrix may have a plurality of sub-matrices such that each submatrix is cyclically shifted version of an identity matrix of size (z×z), where z=162, for example. A parity check matrix can be represented in two equivalent forms: (1) parity check matrix H and (2) a block matrix or an exponent matrix P=E(H).

[0052] In some implementations, a parity check matrix H may be a binary matrix whose size is m×n (each of m and n is an integer). Elements of the parity check matrix are binary values. Given a block length n and a code rate R, an LDPC code (or QC-LDPC code) LDPC (n, R) satisfies the following equations:

$$k = nR \qquad \text{(Equation 3)}$$

$$m = n(1 - R) \qquad \text{(Equation 4)}$$

[0053] In some implementations, a block matrix or an exponent matrix (QC-LDPC exponent matrix) may be obtained. Given a lifting size z, the exponent matrix P=E(H) may have a size of m/z×n/z. If n=24z (e.g., n=3888, z=162), then the size of P=E(H) is 24(1−R)×24 (=n(1−R)/z×n/z). Elements of the exponent matrix may be integer values which correspond to cyclic shift values of identity matrix of size z×z. A parity check matrix H may be a sparse binary matrix that can be derived from an exponent matrix P-E(H). The generator matrix G may have a size n×k in binary form (e.g., elements of the generator matrix G are binary values). The exponent matrix P=E(H) may have a structure including a plurality of sub-matrices (e.g., A, B, C, D, E, T).

[0054] In some implementations, a binary QC-LDPC code LDPC (n, R) may be characterized by the null space of an n(1−R)×n parity check matrix H. The parity check matrix H may be a binary sparse matrix which includes a set of circulant matrices of size z×z. The parity-check matrix H of a QC-LDPC code can be represented equivalently by an exponent matrix P=E(H). This representation can help to illustrate the graphical structure of the underlying code as a base graph along with coefficient of shifting.

[0055] In some implementations, a parity check matrix H may be generated from an exponent matrix P=E(H). The exponent matrix P-E(H) may include (as elements) shift values d in the range 0<=d<z along with d=−1. For example, if z=7, the shift values d may include −1, 0, 1, 2, 3, 4, 5, 6. The shift value d=0 may correspond (or map) to an identity matrix of size z×z, denoted by I(z). The shift value d=−1 may correspond (or map) to a null matrix (all elements zero) of size z×z, denoted by 0*I(z). Any other integer value d in [1,z−1] may correspond (or map) to a matrix cyclically right shifted from I(z). The parity check matrix H can be obtained from the exponent matrix P=E(H) by expanding the exponent matrix P such that each element of the exponent matrix P (as a shift value d) is replaced by a matrix corresponding to the shift value.

[0056] In some implementations, the exponent matrix P=E(H) may include a plurality of elements P1,1, P1,2, P1,3, . . . , P1,ń; P2,1, P2,2, P2,3, . . . , P2,ń; . . . , P1, 1, P1,2, P1,3, . . . , Pḿ,ń, which correspond to (ḿ×ń) values where m and n satisfy the following equations:

$$\acute{m} = n(1 - R)/z \qquad \text{(Equation 5)}$$

$$\acute{n} = n/z \qquad \text{(Equation 6)}$$

[0057] The exponent matrix (or permutation matrix) P=E (H) may be expressed as following:

$$P \equiv E(H) = \begin{pmatrix} P_{1,1} & P_{1,2} & P_{1,3} & \cdots & \cdots & P_{1,\acute{n}} \\ P_{2,1} & P_{2,2} & P_{2,3} & \cdots & \cdots & P_{2,\acute{n}} \\ \cdots & \ddots & \ddots & \cdots & \cdots & \vdots \\ P_{\acute{m},1} & P_{\acute{m},2} & P_{\acute{m},3} & \cdots & \cdots & P_{\acute{m},\acute{n}} \end{pmatrix} \qquad \text{(Equation 7)}$$

[0058] The corresponding parity check matrix H may be obtained by replacing each element of the matrix (as a shift value d) by a matrix C(d) corresponding to the shift value as follows:

$$H = \begin{pmatrix} C(P_{1,1}) & C(P_{1,2}) & C(P_{1,3}) & \cdots & \cdots & C(P_{1,\acute{n}}) \\ C(P_{2,1}) & C(P_{2,2}) & C(P_{2,3}) & \cdots & \cdots & C(P_{2,\acute{n}}) \\ \cdots & \ddots & \ddots & \cdots & \cdots & \vdots \\ C(P_{\acute{m},1}) & C(P_{\acute{m},2}) & C(P_{\acute{m},3}) & \cdots & \cdots & C(P_{\acute{m},\acute{n}}) \end{pmatrix} \quad \text{(Equation 8)}$$

[0059] For example, a matrix C(1) may be expressed as follows:

$$C(1) = \begin{pmatrix} 0 & 1 & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 1 & \cdots & \cdots & \ddots \\ \ddots & \ddots & \ddots & \cdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \cdots & 1 \\ 1 & 0 & 0 & \cdots & \cdots & 0 \end{pmatrix} \quad \text{(Equation 9)}$$

[0060] In some implementations, an encoder can produce codewords using a generator matrix (e.g., using Equation 2). In some implementations, an encoder can use the parity check matrix (rather than the generator matrix) to produce codewords from vectors of information bits. After a parity check matrix H is obtained, the parity check matrix H may have sub-matrices A, B, C, D, T, E. An upper area O of the sub-matrix T may correspond to an area in which the matrix contains all zeroes, and the other areas may represent locations that can contain ones.

[0061] In some implementations, the codeword c can be obtained by the following expression:

$$c = [\,s \quad p1 \quad p2\,], \quad \text{(Equation 10)}$$

[0062] where "s" is the vector of information bits to be encoded, "p_1" is a vector of the first g parity bits and "p_2" is a vector of the remaining m−g parity bits.

[0063] The vectors p1 and p2 can be obtained by the following equations:

$$\Phi = -ET^{-1}B + D; \quad \text{(Equation 11)}$$

$$p_1^T = -\Phi^{-1}(-ET^{-1}A + C)s^T; \quad \text{(Equation 12)}$$

and

$$p_2^T = -T^{-1}(As^T + Bp_1^T). \quad \text{(Equation 13)}$$

[0064] Although various embodiments disclosed herein are described for encoding data for a wireless communication (e.g., wireless local area network (WLAN) conforming to any IEEE 802.11 standard), principles disclosed herein are applicable to other types of communication (e.g., wired communication) or any process that performs encoding for LDPC codes.

[0065] FIG. **3** is a diagram depicting an example exponent matrix (QC-LDPC exponent matrix) **300**, according to one or more embodiments. Given a lifting size z, the exponent matrix **300** may have a size of m/z×n/z. If n=24z (e.g., n=3888, z=162), then the size of P=E(H) is 24(1−R)×24 (=n(1−R)/z×n/z). Elements of the exponent matrix may be integer values which correspond to cyclic shift values of identity matrix of size z×z. A parity check matrix H (see FIG. **5**) may be a sparse binary matrix that can be derived from an exponent matrix P=E(H). The generator matrix G may have a size n×k in binary form (e.g., elements of the generator matrix G are binary values). Referring to FIG. **3**, the exponent matrix P=E(H) may have a structure including a plurality of sub-matrices (e.g., A **310**, B **312**, C **316**, D **318**, E **320**, T **314**).

[0066] FIG. **4** is a diagram **400** depicting example shifted identity matrices **409**, **410**, **411**, **412**, **413**, **414**, **415**, **416** for generating a parity check matrix, according to one or more embodiments. A parity check matrix H may be generated from an exponent matrix P=E(H) (e.g., exponent matrix **300**) or may be identified using a codebook. As shown in Equation 7, the exponent matrix P=E(H) may include (as elements) shift values d in the range 0<=d<z along with d=−1. See Equation For example, if z=7, the shift values d may include −1, 0, 1, 2, 3, 4, 5, 6 (see FIG. **4**). The shift value d=0 may correspond (or map) to an identity matrix of size z×z, denoted by I(z) (e.g., matrix **410**). The shift value d=−1 may correspond (or map) to a null matrix (all elements zero) of size z×z, denoted by 0*I(z) (e.g., matrix **409**). Any other integer value d in [1,z−1] may correspond (or map) to a matrix cyclically right shifted from I(z) (e.g., matrices **411**, **412**, **413**, **414**, **415**, **416**). As shown in Equation 8, the parity check matrix H can be obtained from the exponent matrix P=E(H) by expanding the exponent matrix P such that each element of the exponent matrix P (as a shift value d) is replaced by a matrix corresponding to the shift value.

[0067] FIG. **5** is a diagram depicting an example parity check matrix **500**, according to one or more embodiments. In some implementations, an encoder (e.g., encoder **130**) can produce codewords using a generator matrix (e.g., using Equation 2). In some implementations, an encoder (e.g., encoder **130**) can use the parity check matrix (rather than the generator matrix) to produce codewords from vectors of information bits. After a parity check matrix H is obtained (e.g., using a codebook), the parity check matrix H (e.g., parity check matrix **500**) may have sub-matrices A **510**, B **512**, C **516**, D **518**, T **514**, E **520**. An upper area O **515** of the sub-matrix T **514** (e.g., white area in FIG. **5**) may correspond to an area in which the matrix contains all zeroes, and the other areas (e.g., grey area in FIG. **5**) may represent locations that can contain ones. The size of the parity check matrix **500** may be m×n where the size of the sub-matrix D **518** is g×g, and the size of the sub-matrix T is (m−g)×(m−g). In some implementations, given a vector s of information bits to be encoded, the encoder can obtain a codeword c using Equation 10, Equation 11, Equation 12 and Equation 13.

[0068] FIG. **6** is a diagram depicting an example exponent matrix (QC-LDPC exponent matrix) **600**, according to one or more embodiments. Given a lifting size z, the exponent matrix **600** may have a size of m/z×n/z. If n=48z (e.g., n=3888, z=81), then the size of P=E(H) matrix for n=3888 is 48(1−R)×48 (=n(1−R)/z×n/z). Elements of the exponent matrix may be integer values which correspond to cyclic shift values of identity matrix of size z×z. A parity check matrix H may be a sparse binary matrix that can be derived from an exponent matrix P=E(H). The generator matrix G may have a size n×k in binary form (e.g., elements of the generator matrix G are binary values). Referring to FIG. **6**, the exponent matrix P=E(H) may have a structure including a plurality of sub-matrices (e.g., Á**310**, B **312**, Ć **316**, D **318**, É **320**, Ť **314**).

[0069] FIG. **7**A, FIG. **7**B, and FIG. **7**C are diagrams **700**, **740**, **780** depicting example rate matching schemes including shortening, puncturing, and repeating, respectively, according to one or more embodiments. Referring to FIG. **4**A, a system (e.g., communication system **105**, baseband circuitry **110**, encoder **130**) may zero pad data bits (e.g., zero padded bits or "shortened bits" **701**) to match the number of systematic bits per codeword (e.g., systematic bits **702**) before encoding, and discard the shortened bits **701** after encoding. In this manner, the bit shortener can reduce the effective code rate and improve a coding gain.

[0070] Referring to FIG. **7**B, the system may discard some parity bits (e.g., "punctured bits" **741**) after encoding. In this manner, the system can increases the effective code rate and degrade a coding gain. Referring to FIG. **7**C, the system may copy some bits (e.g., "repeated bits" **781**) from the start of the codeword. In this manner, the system can to improves SNR.

[0071] Aspects of the operating environments and components described above will become apparent in the context of the systems and methods disclosed herein.

C. Systems and Methods for Achieving Extended Long Range in WLANs

[0072] In one aspect, generally, in WLAN systems, access points (APs) transmit at higher power (using an external power amplifier) compared to non-AP stations (STAs), which typically use more efficient power amplifiers or in some cases integrated complementary metal-oxide semiconductor (CMOS) power amplifiers. This may result in a link budget difference of several dB (3-6 dB) between the downlink and uplink communications. This situation (e.g., link budget difference) may exist due to various reasons, such as increased size in AP chipsets and components (e.g., high power amplifier), which can add more sophistication compared to client devices (which are often smaller and have stricter form factor restrictions). The uplink range can be enhanced to address the imbalance in link budget between the downlink (AP to STA, e.g., router to laptop/phone) and uplink (STA to AP). Also in some countries, the regulatory allows APs to transmit significantly higher power levels and or with a higher power spectral density (dBm/MHz) than non-AP STAs, which may create an imbalance between uplink and downlinks.

[0073] To solve the problem, systems and methods of the present disclosure can implement extended long range communication to achieve an aggregate throughput in multiple of 1 Mbps data rate (or multiples of 1 Mpbs, 1.1 Mbps, 1.2 Mbps, 1.5 Mbps, 1.7 Mbps) in a WLAN, thereby reducing an link budget difference or an imbalance between uplink and downlinks. Here, data rate or PHY data rate refers to bit per second (bps), kilobit per second (Kbps), megabit per second (Mbps), gigabit per second (Gbps), or an amount of data transported through a network in a certain period (e.g., second). In some implementations, a system can provide a packet communication mode by which a maximum PHY data rate is substantially equal to or close to 1 Mpbs, 1.1 Mbps, 1.2 Mbps, 1.5 Mbps, 1.7 Mbps or their multiples by means of 802.11 packet transmission technology.

[0074] In some implementations, in order to achieve extended (or enhanced) long range communication, communication systems in communication standards (e.g., IEEE 802.11bn UHR chips) can provide a mode to support PHY rate as low as 1 Mpbs. In some implementations, a system can provide a mode to support extended range with data rates (e.g., PHY rates) as low as 1 Mbps to achieve extended long range communication in WLANs (e.g., 802.11bn) and allow for the least airtime to transmit (e.g., take the least bandwidth (BW) in order to improve efficiency) and minimal processing. For example, communication systems in a WLAN (e.g., IEEE 802.11bn UHR chips) can provide a mode to support PHY rate as low as 1 Mpbs. In some implementations, systems and methods can provide a mode to achieve an aggregate throughput in 1 Mpbs, 1.1 Mbps, 1.2 Mbps, 1.5 Mbps, 1.7 Mbps or their multiples.

[0075] In some implementations, in order to achieve extended long range communication, systems and methods can provide two different designs/schemes/configurations depending on the forward error correction (FEC) scheme: (1) LDPC and (2) BCC (binary convolutional codes). Each of LDPC and BCC can have a corresponding code rate R. The code rate R refers to a ratio of k bits of information bits to n coded bits such that a block of k bits of information bits can be encoded to produce n coded bits (or codeword or coded block). BCC refers to an error-correcting code that generates parity symbols by applying a Boolean function (e.g., sliding Boolean polynomial function) to a data stream. For example, a BCC with a code rate (R) of 1/3 and a length (L) of 7 can generate parity symbols by applying a Boolean polynomial function represented by [171 145 133].

[0076] In some implementations, systems and methods can provide a long range transmission design/configuration/structure with LDPC codes. The systems and methods can generate/provide/create LDPC codes with a code rate R=1/3 using quadrature phase shift keying (QPSK) as a modulation scheme. The systems and methods can use RU/dRU (distributed RU or interleaved RU) and/or LTF (long training field)/CP (cyclic prefix). In some implementations, a rotated QPSK modulation can be used with RU (e.g., RU26 containing 24 data tones and 2 pilot tones). The rotated QPSK ($\pi/4$-QPSK) modulation refers to a variation of QPSK modulation in which the constellation at each symbol is rotated by $\pi/4$ radians from the previous symbol. In some implementations, RU selection can be performed based on existing standards (e.g., the IEEE 802.11ax/be) and/or based on dRU. The systems and methods can use 4×LTF and guard interval (CP=3.2 μs). The guard interval (referred to as "GI") refers to a time interval inserted between successive symbols in a transmitted signal to mitigate the effects of multipath fading. The rate 1/3 based LDPC codes can be chosen either as a native standalone code and/or can be a derived code derived from codes with other rates. For example, from a rate 1/2 LDPC code, a system can derive/construct/create a shortened code of matching rate=1/3. Similarly, from a low rate LDPC code such as a rate of 1/4, the system can derive/construct/create a rate 1/3 code using techniques such as puncturing.

[0077] In some implementations, systems and methods can provide a long range transmission design/configuration/structure with BCC codes. In some implementations, the systems and methods can generate/provide/create BCC codes with a code rate R=2/3 using BPSK as a modulation scheme. RU/dRU and LTF/CP can be used similar to that used in LDPC. For example, the systems and methods can use 4×LTF and guard interval (CP=3.2 us). In some implementations, RU selection can be performed based on existing standards (e.g., the IEEE 802.11ax/be) and/or based on dRU. In some implementations, the systems and methods can use a rate 1/3 BCC with a QPSK modulation scheme. The systems and methods can use/generate/provide/create/design 1/3 BCC codes based on existing 1/3 trellis based convolutional codes, or alternatively, by concatenating a rate 2/3 BCC code with 1/2 repetition code. The concatenation can be performed using 2/3 BCC as an inner code or as an outer code (e.g., the repetition code can swap the order in each case). If the code is used/generated/provided/created/designed as a serial concatenation, the systems and methods can optionally place an interleaver between the constituent codes.

[0078] In some implementations, when a RU26 (or 26-distributed RU or interleaved RU (26-dRU)) including 24 information tones (or data tones) and **2** pilot tones is used, a physical data rate PHY_RATE_{26} can be calculated as follows:

$$PHY\_RATE_{26} = R * \log_2 M * 24/(4*3.2 + GI), \quad \text{Equation (14)}$$

[0079] where R refers to a code rate of an FEC scheme (e.g., LDPC or BCC); M is the number of phases used in a phase-shift keying (PSK) scheme, e.g., **2** for BPSK, **4** for QPSK, etc.; and GI is a guard interval.

[0080] When RU26s are used, the data rate PHY_RATE_{26} can be calculated using Equation 14. For example, if M=4 (QPSK), R=1/3, GI=3.2 μs, then PHY_RATE_{26}=1/3*log 2(4)*24/(4*3.2+3.2)=1 Mbps; if M=2 (BPSK), R=2/3, GI=3.2 μs, then PHY_RATE_{26}=2/3*log 2(2)*24/(4*3.2+3.2)=1 Mbps; if M=2 (BPSK), R=2/3, GI=1.6 μs, then PHY_RATE_{26}=2/3*log 2(2)*24/(4*3.2+1.6)=1.11 Mbps; if M=4 (QPSK), R=1/3, GI=1.6 μs, then PHY_RATE_{26}=1/3*log 2(4)*24/(4*3.2+1.6)=1.11 Mbps; if M=2 (BPSK), R=2/3, GI-0.8 μs, then PHY_RATE_{26}=2/3*log 2(2)*24/(4*3.2+0.8)=1.176 Mbps; if M=4 (QPSK), R=1/3, GI-0.8 μs, then PHY_RATE_{26}=1/3*log 2(4)*24/(4*3.2+0.8)=1.176 Mbps. More examples can be shown in Table 1 and Table 2 below. In some implementations, in a given PHY bandwidth (e.g., 24 MHz bandwidth), one or more such RU/dRU (e.g., RU26/26-dRU) may be repeated for additional diversity gain.

[0081] In some implementations, when a RU52 (or 52-dRU) including 48 information tones (or data tones) and **4** pilot tones is used, a physical data rate PHY_RATE_{52} can be calculated as follows:

$$PHY\_RATE_{52} = R * \log_2 M * 48/(4*3.2 + GI), \quad \text{Equation (15)}$$

[0082] where R refers to a code rate of an FEC scheme (LDPC or BCC); M is the number of phases used in a PSK scheme, e.g., 2 for BPSK, 4 for QPSK, etc.; and GI is a guard interval.

[0083] When RU52s are used, the data rate PHY_RATE_{52} can be calculated using Equation 15. For example, if M=2 (BPSK), R=1/2, GI-3.2 μs, then PHY_RATE_{52}=1/2*log 2(2)*48/(4*3.2+3.2)=1.5 Mbps; if M=2 (BPSK), R=1/2, GI=1.6 μs, then PHY_RATE_{52}=1/2*log 2(2)*48/(4*3.2+1.6)=1.67 Mbps; if M=2 (BPSK), R=1/2, GI-0.8 μs, then PHY_RATE_{52}=1/2*log 2(2)*48/(4*3.2+0.8)=1.76 Mbps. More examples can be shown in Table 1 and Table 2 below. In some implementations, in a given PHY bandwidth (e.g., 24 MHz bandwidth), one or more such RU/dRU (e.g., RU52/52-dRU) may be repeated for additional diversity gain.

[0084] In some implementations, systems and methods can provide a long range transmission design/configuration/structure using multiple-input and multiple-output (MIMO) systems and/or orthogonal frequency-division multiple access (OFDMA) modulations. In some implementations, systems and methods can use MIMO optionally, or can be configured/generalized for arbitrary selection of antennas, to implement long range transmission.

[0085] In some implementations, the systems and methods can achieve the target rate of PHY rate (also referred to as "target data rate" or "target PHY rate") to be equal to 1 Mbps with a 1 spatial stream (e.g., the number of spatial streams Nss=1). Here, "target data rate" or "target PHY rate" refers to a data rate or a PHY rate that is to be achieved as a result of a particular long range transmission design/configuration/structure. The target data rate or target PHY rate can be achieved based on any of the following antenna configurations: (1) 1 transmit, 1 receive antennas; (2) (Ntx, Nrx) MIMO where Ntx, Nrx refer to the number of transmit antennas and the number of receive antennas, respectively, and min (Ntx, Nrx)=1; and (3) If min (Ntx, Nrx)=d, then the achievable rate=d*1 Mbps can be realized with the same design/configuration (e.g., LDPC configuration to achieve 1 Mbps). For example, if a 2×4 MIMO configuration (e.g., d=2) with or without beamforming can realize 2 Mbps rate.

[0086] In some implementations, the systems and methods can adjust the number of antennas to stay or remain similar to the design/configuration with LDPC or BCC. The base design/configuration can achieve 1 Mbps, assumes Nss=1, which can be realized with (Ntx, Nrx) where min (Ntx, Nrx)=1. MIMO can be generalized for arbitrary selection of antennas. MIMO can provide additional diversity/combining gain depending on the configurations. For example, if a 1×2 MIMO configuration can provide 3 dB, then a 1×4 MIMO configuration can provide 6 dB etc.

[0087] In some implementations, the base RU can be 26 or 52 (e.g., RU26 or RU52) depending on the configuration. In a given PHY bandwidth, a number of such RUs can be repeated (e.g., in 20 MHz PHY bandwidth, potentially up to 9 such RU26 can be repeated). Such repetition offers diversity gain when there is multipath fading in the channel. RU52 can be similarly repeated 2 to 4 times in 20 MHz.

[0088] In some implementations, the LDPC code with a code rate R=1/3 can be a native code or can be obtained by one of the following schemes: (1) shortening from a 1/2 LDPC code; (2) concatenating a 2/3 LDPC code with repetition code of rate 1/2; (3) a blockwise repetition with a 2/3 LDPC code; (4) bit-level repetition with 2/3 LDPC coded bits with or without interleaving; or (5) 1/2 coded bits followed by a combination of shortening, repetition and puncturing to obtain an effective code rate=1/3. In some implementations, the system can adjust the number of shortened/padding/repeated/punctured bits for a given payload length.

[0089] In some implementations, by using higher quadrature amplitude modulation(QAM), for example, as the number M in M-ary QAM increases, a proportional increase in data rate can be achieved (at the cost of higher SNR and thus reduced range). M-ary QAM refers to different constellation sizes (M) in the constellation diagram. Examples include 16-QAM, 64-QAM, and 256-QAM.

[0090] Table 1 below shows various configurations to achieve target data rates using LDPC codes.

TABLE 1

Summary of Configurations Using LDPC Codes

Target Data Rate	RU/ dRU	Code Rate	Native/Shorten/Puncture	Modulation (M)	Guard Interval (GI)	Result Data Rate
1 Mbps	RU26	1/3	Native	4 (QPSK)	3.2	1 Mbps
1 Mbps	RU26	1/3	Shortened using 2/3 code	4 (QPSK)	3.2	1 Mbps
1 Mbps	RU26	1/3	Punctured using 1/4 code	4 (QPSK)	3.2	1 Mbps
1 Mbps	RU26	2/3	Native	2 (BPSK)	3.2	1 Mbps
1 Mbps	RU26	2/3	Native	2 (BPSK)	1.6	1.11 Mbps
1 Mbps	RU26	2/3	Native	2 (BPSK)	0.8	1.176 Mbps
1 Mbps	RU26	1/3	Native	4 (QPSK)	1.6	1.11 Mbps
1 Mbps	RU26	1/3	Shortened using 2/3 code	4 (QPSK)	1.6	1.11 Mbps
1 Mbps	RU26	1/3	Punctured using 1/4 code	4 (QPSK)	1.6	1.11 Mbps
1 Mbps	RU26	1/3	Native	4 (QPSK)	0.8	1.176 Mbps
1 Mbps	RU26	1/3	Shortened using 2/3 code	4 (QPSK)	0.8	1.176 Mbps
1 Mbps	RU26	1/3	Punctured using 1/4 code	4 (QPSK)	0.8	1.176 Mbps
1.5 Mbps	RU52	1/2	Native	2 (BPSK)	3.2	1.5 Mbps
1.5 Mbps	RU52	1/2	Native	2 (BPSK)	1.6	1.67 Mbps
1.5 Mbps	RU52	1/2	Native	2 (BPSK)	0.8	1.76 Mbps

[0091] Table 2 below shows various configurations to achieve target data rates using BCC codes.

TABLE 2

Summary of Configurations Using BCC Codes

Target Data Rate	RU/ dRU	Code Rate	Native/Repeated	Modulation (M)	Guard Interval (GI)	Result Data Rate
1 Mbps	RU26	1/3	Native	4 (QPSK)	3.2	1 Mbps
1 Mbps	RU26	1/3	A native 2/3 code concatenated with a 1/2 repetition code	4 (QPSK)	3.2	1 Mbps
1 Mbps	RU26	2/3	Native	2 (BPSK)	3.2	1 Mbps
1 Mbps	RU26	2/3	Native	2 (BPSK)	1.6	1.11 Mbps
1 Mbps	RU26	2/3	Native	2 (BPSK)	0.8	1.176 Mbps
1 Mbps	RU26	1/3	Native	4 (QPSK)	1.6	1.11 Mbps
1 Mbps	RU26	1/3	A native 2/3 code concatenated with a 1/2 repetition code	4 (QPSK)	1.6	1.11 Mbps
1 Mbps	RU26	1/3	Native	4 (QPSK)	0.8	1.176 Mbps
1 Mbps	RU26	1/3	A native 2/3 code concatenated with a 1/2 repetition code	4 (QPSK)	0.8	1.176 Mbps
1.5 Mbps	RU52	1/2	Native	2 (BPSK)	3.2	1.5 Mbps
1.5 Mbps	RU52	1/2	Native	2 (BPSK)	1.6	1.67 Mbps
1.5 Mbps	RU52	1/2	Native	2 (BPSK)	0.8	1.76 Mbps

[0092] In some implementations, an apparatus may include a transmitter and one or more processors. The one or more processors may be configured to identify a target data rate for transmitting data over a channel with a frequency bandwidth. Based at least on the target data rate, the one or more processors may be configured to select a forward error correction (FEC) code, a code rate, a modulation scheme, and a number of resource units (RUs) within the frequency bandwidth, to transmit the data within a range of the target data rate. The one or more processors may be configured to encode, by an FEC encoder, the data using the FEC code and the code rate to generate encoded data. The one or more processors may be configured to modulate the encoded data using the modulation scheme to generate modulated data. The transmitter may be configured to transmit the modulated data using the number of RUs.

[0093] In some implementations, an apparatus may include a transmitter and one or more processors. The one or more processors may identify a target data rate for transmitting data over a channel with a frequency bandwidth. Based at least on the target data rate, the one or more processors may select a forward error correction (FEC) code, a code rate, a modulation scheme, and a number of resource units (RUs) within the frequency bandwidth, to transmit the data within a range of the target data rate. The one or more processors may encode, by an FEC encoder, the data using the FEC code and the code rate to generate encoded data. The one or more processors may modulate the encoded data using the modulation scheme to generate modulated data. The transmitter may transmit the modulated data using the number of RUs.

[0094] In some implementations, the selected FEC code may be a low density parity check (LDPC) code or a binary convolutional (BCC) code. In some implementations, based at least on the target data rate, the one or more processors are further configured to select a guard interval (GI) and a number of tones per RU, to transmit the data within a range of the target data rate. In some implementations, the target data rate may be 1 Mbps or 1.5 Mbps. The range of the target data rate may be 20% of the target data rate.

[0095] In some implementations, the FEC code may have a first code rate that is different from the selected code rate. The one or more processors may be configured to encode, by the FEC encoder, the data using the FEC code with the first code rate to generate encoded data that corresponds to the selected code rate.

[0096] In some implementations, the FEC code may be an LDPC code, the selected code rate may be 1/3, and the first code rate may be ½ or 1/4. In some implementations, the FEC code may be a BCC code, the selected code rate may be 1/3, and the first code rate may be 2/3.

[0097] Embodiments in the present disclosure have at least the following advantages and benefits. First, embodiments in the present disclosure can provide useful techniques for reducing an link budget difference or an imbalance between uplink and downlinks. In some implementations, a system can provide a packet communication mode by which a maximum PHY data rate is substantially equal to or close to 1 Mpbs, 1.1 Mbps, 1.2 Mbps, 1.5 Mbps, 1.7 Mbps or their multiples by means of 802.11 packet transmission technology.

[0098] Second, embodiments in the present disclosure can provide useful techniques for achieving target data rates using variable schemes of encoding, modulation, and/or transmission. Table 1 and Table 2 show a summary of such schemes to achieve target data rates.

[0099] Third, embodiments in the present disclosure can provide useful techniques for reusing existing codes to help pave the way for similar design. For example, from a rate 1/2 LDPC code, a system can derive/construct/create a shortened code of matching rate=1/3. Similarly, from a low rate LDPC code such as a rate of 1/4, the system can derive/construct/create a rate 1/3 code using techniques such as puncturing. The systems and methods can use/generate/provide/create/design 1/3 BCC codes based on existing 1/3 trellis based convolutional codes, or alternatively, by concatenating a rate 2/3 BCC code with 1/2 repetition code. The concatenation can be performed using 2/3 BCC as an inner code or as an outer code (e.g., the repetition code can swap the order in each case).

[0100] FIG. **8** is a diagram depicting a structure of an ELR packet **850** and various examples of schemes **810**, **820**, **830**, **840** for encoding, modulating and transmitting the ELR packet **850** to obtain a target data rate, according to one or more implementations. The ELR packet **850** may be a physical layer protocol data unit (PPDU) including a legacy preamble **851**, an ELR preamble **852**, and a data payload **853**. The legacy preamble **851** may be a WLAN-compatible (e.g., IEEE 802.11be-compatible) preamble. The ELR preamble **852** may be used for detection of ERL packets (e.g., by containing a particular sequence of bit values). The data payload **853** may be processed to achieve a target data rate (e.g., target rate of 1 Mbps or within a 20% range of 1 Mbps) using various schemes. Such schemes may include a scheme **810**, a scheme **820**, a scheme **830**, and a scheme **840**. The scheme **810** may include encoding using a forward error correction (FEC) code with a data rate of 2/3 (**810-1**), concatenating the FEC code with a repetition code with a data rate of 1/2 (**810-2**), modulating using QPSK (**810-3**), channel modulating using (**810-4**), OFDM and/or transmitting using the defined/allocated/designed/configured RU26 or dRU26 (**810-5**) over an OFDMA channel. The scheme **820** may include encoding using an FEC code with a data rate of 2/3 (**820-1**), concatenating the FEC code with a repetition code with a data rate of 1/2 (**820-2**), modulating using BPSK (**820-3**), channel modulating using OFDM (**820-4**), and/or transmitting using the defined/allocated/designed/configured RU52 or dRU52 (**820-5**) over an OFDMA channel. The scheme **830** may include encoding using an FEC code with a data rate of 1/3 (**830-2**), modulating using BPSK (**830-3**), channel modulating using OFDM (**830-4**), and/or transmitting using the defined/allocated/designed/configured RU52 or dRU52 (**830-5**) over an OFDMA channel. The scheme **840** may include encoding using an FEC code with a data rate of 1/3 (**840-2**), modulating using QPSK (**840-3**), channel modulating using OFDM (**840-4**), and/or transmitting using the defined/allocated/designed/configured RU26 or dRU26 (**840-5**) over an OFDMA channel.

[0101] FIG. **9** is a diagram depicting a structure of an ELR packet **950** and an example of schemes **910** for encoding, modulating and transmitting the ELR packet **950** to obtain a target data rate, according to one or more implementations. The ELR packet **950** may be a PPDU including a legacy preamble **951**, an ELR preamble **952**, and a data payload **953**. The legacy preamble **951** may be a WLAN-compatible (e.g., IEEE 802.11be-compatible) preamble. The ELR preamble **952** may be used for detection of ERL packets (e.g., by containing a particular sequence of bit values). The data payload **953** may be processed to achieve a target data rate (e.g., target rate of 1.5 Mbps or within a 20% range of 1 Mbps) using various schemes, e.g., scheme **910**. The scheme **910** may include encoding using an FEC code with a data rate of 1/2 (**910-2**), modulating using BPSK (**910-3**), channel modulating using OFDM (**910-4**), and/or transmitting using the defined/allocated/designed/configured RU52 or dRU52 (**910-5**) over an OFDMA channel.

[0102] FIG. **10** is a diagram depicting an allocation structure **1000** of resource units (RUs) in a bandwidth of 20 MHz, according to one or more implementations. A 20 MHz channel can be divided into multiple RUs, each including a number of subcarriers (or tones). For example, 256 tones of the 20 MHz chancel can be allocated/grouped into (1) RUs including 26 tones (referred to as "RU26s"), (2) RUs including 52 tones (referred to as "RU52s"), or (3) RUs including 106 tones (referred to as "RU106s"). FIG. **4** shows different RU allocation structures **1010**, **1020**, **1030**, **1040**, **1050** and **1060** in a bandwidth of 20 MHz. For example, the RU allocation structure **1010** may include 9 RU26s (e.g., RU26 **1012**); the RU allocation structure **1020** may include 6 RU26s and a combination of RU26 and RU52 (e.g., RU **1022**); the RU allocation structure **1030** may include 6 RU26s and a combination of RU52 and RU26 (e.g., RU **1032**); the RU allocation structure **1040** may include 4 RU26s and a combination of RU106 and RU26 (e.g., RU **1042**); the RU allocation structure **1050** may include 4 RU26s and a combination of RU106 and RU26 (e.g., RU **1052**).

[0103] FIG. **11** is a diagram depicting a structure **1100** of repeating RUs in a bandwidth of 20 MHz, according to one or more implementations. A system can repeat an RU (e.g., RU52 **1101**) four times to produce repeated RUs **1103** which is a vector having four RU52s as elements. In some implementations, a vector HR **1111** can be defined such that each element of the vector represents a phase shift of input signal (e.g., input signal corresponding to an RU52). For example, $H_R=[+1\ -1\ -1\ +1]$ may represents phase shifts of +90°, −90°, −90°, +90°. In some implementations, a system may receive a stream of vectors H_R **1112**, select one of the stream of vectors H_R **1112** (e.g., H_R **1113**) and perform a Hadamard product (e.g., element-wise multiplication) on the selected H_R (e.g., H_R **1113**) and the RU52 vector **1103**. As a result of the Hadamard product, the system may generate an output **1130** which includes RU52-1 (**1131**), RU52-2 (**1132**), RU52-3 (**1133**), and RU52-4 (**1134**) which correspond to the RU52 (RU52 **1101**) phase shifted by +90°, −90°, −90°, +90°, respectively.

[0104] FIG. **12** is a diagram depicting an example encoder **1200** for encoding with a shortened LDPC code to obtain a particular code rate (e.g., target code rate r), according to one or more embodiments. The encoder **1200** may include an LDPC encoder **1240**, an information bits optimizer **1220**, and/or a MUX **1260**. The encoder **1200** and each component thereof may be implemented using circuitry, firmware and/or software. For example, the encoder **1200** may be implemented in the encoder **130** of the communication system **105**. The LDPC encoder **1240** may take a block of k bits of information bits and produce n coded bits (or codeword or coded block) with the code rate R=k/n.

[0105] Referring to FIG. **12**, the information bits optimizer **1220** may include a bit permutation pattern or permutation pattern **1222** (denoted by Π) and/or masked bits **1224** (or bitmask). The permutation pattern **1222** may define or indicate how input bits can be permutated. The permutation pattern **1222** and/or the masked bits **1224** may be chosen a priori. The permutation pattern **1222** and/or the masked bits **1224** may be stored in memory. Given k information bits, the permutation pattern **1222** may define, indicate, or store an index to one permutation of all permutations of k information bits (e.g., an index to one permutation of all k! permutations). For example, the permutation pattern **1222** may indicate one of 1, 2, . . . , k!, each of which indicates a permutation of k information bits. The bitmask **1224** may be an array of bits whose size is equal to k information bits such that each bit of the bitmask indicates whether to keep or clear the corresponding bit of the k information bits. The information bits optimizer **1220** may be configured to perform bitwise or bit operations (e.g., NOT, AND, OR, XOR, shift, swap, etc.) to perform permutation(using the permutation pattern Π) and/or bitmasking (using the bitmask) on the k information bits.

[0106] The encoder **1200** may obtain data of coded bit error rate (BER) and/or frame error probability (FEP; from past measurements or simulations for example), and determine, based on the bit error rates and/or frame error probability and prior to generating LDPC codes, the permutation pattern **1222** and/or the bitmask **1224**. Here, the coded bit error rate (BER) refers to a number of coded bit errors per unit time where the number of coded bit errors refers to a number of received coded bits of a data stream that have been altered due to errors (e.g., noise, interference, distortion or any other bit errors). The bit error probability (FEP) refers to an expected value (e.g., mean, or weighted average) of a bit error ratio, where the bit error ratio refers to a number of bit errors divided by the total number of transferred bits during a certain time interval. The encoder **1200** may determine the permutation pattern **1222** to reduce the BER or FEP (e.g., by permuting particular bits that are highly affected by BER or FEP). The encoder **1200** may determine the bitmask **1224** to reduce the BER or FEP (e.g., by clearing particular bits that are highly affected by BER or FEP).

[0107] The encoder **1200** can monitor data streams (e.g., transmission of digitally encoded signals) over a communication channel, determine or measure a coded BER and/or an FEP from the monitored data streams, and use the measured coded BER and/or an FEP to dynamically determine the permutation pattern **1222** and/or the bitmask **1224**. The encoder **1200** can be tunable to set the permutation pattern **1222** and/or the bitmask **1224** to different pattern/bitmasking values.

[0108] The encoder **1200** may receive/determine/identify a desired code rate (or target code rate) denoted by r (e.g., r=1/3) and a (base) code rate of a base LDPC code (e.g., R=1/2). The encoder **1200** may first determine a target code rate r and then determine, based on the target code rate r, a base code rate R (or one or more base code rates). Alternatively, the encoder **1200** may first determine a base code rate R and then determine, based on the base code rate R, a target code rate r (or one or more target code rates).

[0109] The encoder **1200** may determine, based on the base code rate (e.g., R=1/2), values of k and n (e.g., k=822, n=1944). The base LDPC code may have (or be associated with) a code rate R, a codeword length (or coded block length) n, and an information bits size k (e.g., n=648,1296, 1944,3888 if the base code is chosen from the 802.11n-be standards).

[0110] The encoder **1200** may determine a parameter of $\acute{k}$ indicating the size of information bits ($\acute{k}$<k; e. g., $\acute{k}$=486) for the target rate r (e.g., r=1/3). The parameter $\acute{k}$ may be chosen a priori for a target rate r. The encoder **1200** may determine the value of $\acute{k}$ using the following equation:

$$\acute{k} = r * \frac{1-R}{1-r}. \quad \text{(Equation 16)}$$

[0111] The encoder **1200** may determine a parameter of $\acute{n}$ indicating a coded block length (or codeword size; e.g., $\acute{n}$=**1458**) for the target rate r (e.g., r=1/3). The encoder **1200** may determine the value of $\acute{n}$ using the following equation:

$$\acute{n} = n * \frac{1-R}{1-r}. \quad \text{(Equation 17)}$$

[0112] The encoder **1200** may determine a parameter of (k−$\acute{k}$) indicating a frozen block length or a size of one or more bits added to the k information bits (k−$\acute{k}$=486) for the target rate r (e.g., r=1/3). The encoder **1200** may determine the parameter (k−$\acute{k}$) using the following equation:

$$(k - \acute{k}) = \frac{R-r}{1-r}. \quad \text{(Equation 18)}$$

[0113] With the parameters determined above (e.g., $\acute{k}$, $\acute{n}$), the target rate r can be calculated using the following equation:

$$r = \frac{\acute{k}}{\acute{n}} = \frac{\acute{k}}{n - k + \acute{k}}. \quad \text{(Equation 19)}$$

[0114] The encoder **1200** may calculate/compute/obtain possible combinations of parameters (e.g., r, R, ń, ḱ) using Equation 16 to Equation 19 (such that ń, ḱ should be integer values), store the resulting parameter combinations in memory (e.g., in the form of look-up table (LUT)) as shown in Table 1, and determine parameters by selecting a parameter combination from the LUT. Using Table 1, the encoder **1200** can first determine a target code rate r and then determine/select/choose, based on the target code rate r, one or more base code rates R. The encoder **1200** can first determine a base code rate R and then determine, based on the base code rate R, one or more target code rates.

[0115] The encoder **1200** may receive ḱ information bits **1201** for the target code rate r. Next, the encoder **1200** may add (k–ḱ) bits **1202** to the ḱ information bits at the end thereof to form/generate k information bits **1210** for the base code rate R. The added (k–ḱ) bits **1202** may be zero (e.g., (k–ḱ) number of binary zeros). The added (k–ḱ) bits **1202** may include one or more binary ones.

[0116] In response to adding (k–ḱ) bits **1202** to the ḱ information bits **1201**, the information bits optimizer **1220** may receive the resulting k information bits **1210**, perform at least one of permutation using the permutation pattern Π (**1222**) or bitmasking using the bitmask **1224**, and generate optimized k information bits. The encoder **1200** may control the information bits optimizer **1220** to dynamically turn on or off, thereby selectively skipping permutation and/or bitmasking for simplicity without overly compromising the performance of the resulting code. The encoder **1200** may determine a degree of coded BER or FEP is less than a threshold, and turn off the information bits optimizer **1220** to skip permutation and bitmasking.

[0117] In response to the information bits optimizer **1220** generating the optimized k information bits (or in response to adding (k–ḱ) bits **1202** to the ḱ information bits **1201** if the information bits optimization is skipped), the LDPC encoder may receive k information bits, and encode the k information bits to generate k encoded bits (not shown) and (n–k) parity bits **1250**. In response to generating the (n–k) parity bits **1250**, the MUX **1260** may receive, as two inputs, the ḱ information bits **1201** and the (n–k) parity bits **1250**, and generate/output (n–k+ḱ) bits **1280** by concatenating (or combining, merging, integrating, consolidating) the ḱ information bits **1201** and the (n–k) parity bits **1250**. The output (n–k+ḱ) bits **1280** may correspond to a codeword for the LDPC code with target rate r. In other words, the encoder **1200** can encode the ḱ information bits **1201** to generate the (n–k+ḱ) encoded bits **1280** (or a codeword) such that the code rate of the encoder **1200** corresponds to

$$r = \frac{\acute{k}}{n - k + \acute{k}}$$

(Equation 19).

[0118] FIG. **13** is a diagram depicting an example encoder **1300** for encoding with a shortened LDPC code to obtain a particular code rate (e.g., target code rate r=1/3), according to one or more embodiments. Referring to FIG. **13**, the encoder **1300** may receive ḱ information bits **1301** (e.g., **486** information bits), and add (k–ḱ) bits **1302** (e.g., 486 bits **1302**) to the ḱ information bits **1301** to generate k information bits **1310** (e.g., **972** information bits). An LDPC encoder **1340** may receive the **972** information bits **1310** and produce n encoded bits (e.g., 1944 encoded bits) including 972 parity bits **1350**, according to the base code rate R=1/2. A MUX **1360** may concatenate the 486 information bits **1301** and the 972 parity bits **1350** to produce (1458=486+972) encoded bits **1380**. In other words, the encoder **1300** can encode 486 information bits **1301** to generate **1458** encoded bits **1380** (or a codeword) such that the code rate r of the encoder **1300** corresponds to

$$\frac{1}{3} = \frac{486}{1458}.$$

[0119] FIG. **14** is a diagram depicting an example decoder **1400** for decoding with a shortened LDPC code according to one or more embodiments. The decoder **1400** may include an LDPC decoder **1450**, a LLR values optimizer **1420**, a multiplexer (MUX) **1430**, and/or a bits de-optimizer (or inverse optimizer) **1460**. The decoder **1400** and each component thereof may be implemented using circuitry, firmware and/or software. For example, the decoder **1400** may be implemented in the decoder **160** of the communication system **108**. In response to the communication system **105** transmitting encoded bits produced by the encoder **1200**, the communication system **108** may receive the transmitted encoded bits and the decoder **1400** of the communication system **108** may decode LLR values corresponding to the transmitted encoded bits (e.g., LLR values **1410**). The LDPC decoder **1450** may take a block of n encoded bits and produce k decoded bits (or decoded information bits) with the code rate R=k/n.

[0120] Referring to FIG. **14**, the LLR values optimizer **1420** may have configuration similar to the information bits optimizer in the encoder **1200**. For example, the LLR values optimizer **1420** may be configured to perform permutation (using a permutation pattern Π) and/or bitmasking (using a bitmask) on input LLR values (e.g., k LLR values). The bits de-optimizer **1460** may be configured to perform an inverse operation of the information bits optimizer (e.g., information bits optimizer **1220**) in the corresponding encoder (e.g., encoder **1200**).

[0121] The decoder **1400** may receive (n–k+ḱ) log likelihood ratio (LLR) values **1410** including k LLR values **1411** (corresponding to k information bits) and (n–k) LLR values **1412** (corresponding to (n–k) parity bits). The decoder **1400** may obtain/extract, from the ḱ LLR values **1411**, k LLR values **1413** by removing (k–ḱ) LLR values **1414** at the end of the ḱ LLR values **1411**. The removed (k–ḱ) LLR values **1414** may correspond to the (k–ḱ) bits **1202** (e.g., (k–ḱ) zeros) added to the ḱ information bits **1201** by the corresponding encoder **1200**.

[0122] The decoder **1400** may generate, by the LLR values optimizer **1420**, k optimized LLR values. The decoder **1400** may turn off the LLR values optimizer **1420** to skip permutation and bitmasking.

[0123] In response to the LLR values optimizer **1220** generating the optimized k optimized LLR values **1441** (or in response to extracting, from the $\acute{k}$ LLR values **1411**, k LLR values **1413** if the LLR values optimization is skipped), the MUX **1430** may receive, as two inputs, the k (optimized) LLR values **1441** and the (n−k) LLR values **1412**, and generate/output n LLR values **1440** by concatenating (or combining, merging, integrating, consolidating) the k LLR values **1441** and the (n−k) LLR values **1412**.

[0124] The LDPC decoder **1450** may receive the n LLR values **1440** as an output from the MUX **1430**, and decode the n LLR values **1440** to generate k information bits. In response to determining that the corresponding encoder **1200** has performed the information bits optimization, the bits de-optimizer **1460** may be configured to receive the k information bits from the LDPC decoder **1450**, and perform, based on the permutation pattern Π (**1222**) or the bitmask (**1224**) in the encoder **1200**, bitwise or bit operations to perform an inverse permutation and/or inverse bitmasking on input bits (e.g., k information bits). If the corresponding encoder **1200** does not perform the information bits optimization, the decoder **1400** may not perform the inverse bits optimization (e.g., by turning off the bits de-optimizer).

[0125] In some implementations, in response to the bits de-optimizer **1460** generating the de-optimized k information bits **1470** (or in response to the LDPC decoder **1450** generating the k information bits if the corresponding encoder **1200** does not perform the information bits optimization), the decoder **1400** may extract, from the k information bits **1470**, $\acute{k}$ information bits **1471** at the start of the k information bits **1470**, and output the k information bits **1471**. The outputted $\acute{k}$ information bits **1471** may correspond to information bits for the LDPC code with target rate r. In other words, the decoder **1400** can decode the (n−k+$\acute{k}$) LLR values **1410** to generate the $\acute{k}$ information bits **1471** such that the code rate of the decoder **1400** corresponds to

$$r = \frac{\acute{k}}{n - k + \acute{k}}$$

(Equation 19).

[0126] FIG. **15**A and FIG. **15**B are diagrams depicting example simulation results using a system to obtain a target data rate on a channel with (1) LDPC codes, (2) DNLOS signal propagation (e.g., a 802.11 MIMO channel model type D which is non-line of sight) and (3) the number of spatial streams equal to 1 (Nss=1), according to one or more embodiments. Referring to FIG. **15**A, lines **1501**, **1502**, **1503**, **1504** correspond to simulation results (SpecEff vs SNR) using modulation configurations of (1) 2×2 MIMO, QPSK, R=1/3, (2) 2×2 MIMO, BPSK, R=2/3, (3) 1×1 SISO, QPSK, R=1/3, and (4) 1×1 SISO, BPSK, R=2/3, respectively. SpecEff refers to an information rate (or bit rate or effective data rate) over a given bandwidth in a communication system (in the unit of bits/second/Hz). Referring to FIG. **15**B, lines **1551**, **1552**, **1553**, **1554** correspond to simulation results (PER vs SNR) using modulation configurations of (1) 2×2 MIMO, QPSK, R=1/3, (2) 2×2 MIMO, BPSK, R=2/3, (3) 1×1 SISO, QPSK, R=1/3, and (4) 1×1 SISO, BPSK, R=2/3, respectively. As shown in FIGS. **15**A and **15**B, the configuration (1) shows the highest spectral efficiency and the least PER.

[0127] FIG. **16**A is a diagram depicting example simulation results using a system to obtain a target data rate on a channel with (1) LDPC codes, (2) QPSK, and (3) 1×1 SISO, according to one or more embodiments. Referring to FIG. **16**A, lines **1601**, **1602**, **1603**, **1604** correspond to correspond to simulation results (PER vs SNR) using configurations of (1) native LDPC code with R=1/3 on a DNLOS channel, (2) native LDPC code with R=2/3 concatenated with a repetition code with R=1/2 on a DNLOS channel, (3) native LDPC code with R=1/3 on a BLOS channel (e.g., a 802.11 channel model Type B which is line of sight), (4) native LDPC code with R=2/3 concatenated with a repetition code with R=1/2 on a BLOS channel, respectively. As shown in FIG. **16**A, the native codes with R=1/3 have 1 dB or more gain over the concatenation codes.

[0128] FIG. **16**B is a diagram depicting example simulation results using a system to obtain a target data rate on a channel with (1) BCC codes and (2) 1×1 SISO, according to one or more embodiments. Referring to FIG. **16**B, lines **1651**, **1652**, **1653**, **1654** correspond to correspond to simulation results (PER vs SNR) using configurations of (1) native BCC code with R=2/3 concatenated with a repetition code with R=1/2 on a DNLOS, QPSK channel (resulting a code rate of 1/3), (2) native BCC code with R=1/3 on a DNLOS, BPSK channel, (3) native BCC code with R=2/3 concatenated with a repetition code with R=1/2 on a BLOS, QPSK channel (resulting a code rate of 1/3), (4) native BCC code with R=1/3 on a BLOS, BPSK channel, respectively. As shown in FIG. **16**A, the configuration of QPSK with concatenated BCC code (e.g., configuration (1)) and the configuration of BPSK with native BCC code (e.g., configuration (2)) show performance fairly close to each other, although the concatenated BCC code configuration is easier to implement than the native BCC code configuration.

[0129] In some implementations, an apparatus (e.g. communication system **105**) may include a transmitter (e.g., transmitter circuitry **120**) and one or more processors (e.g., processor **2010**). The one or more processors may be configured to identify a target data rate (e.g., 1 Mbps) for transmitting data. Based at least on the target data rate, the one or more processors may be configured to select a modulation scheme and a code rate to transmit the data within a range of the target data rate. For example, the apparatus can select, based on the target data rate, a modulation scheme and a code rate using Table 1 and Table 2. The one or more processors may be configured to identify, based at least on the selected code rate (e.g., code rate of 1/3), a forward error correction (FEC) code (e.g., LDPC code) with a first code rate (e.g., code rate of 1/2; see FIG. **13**) that is different from the selected code rate. The one or more processors may be configured to encode, by an FEC encoder (e.g., LDPC encoder **1340**), the data using the FEC code with first code rate to generate encoded data that corresponds to the selected code rate. The one or more processors may be configured to modulate the encoded data using the modulation scheme (e.g., BPSK or QPSK) to generate modulated data. The transmitter may be configured to transmit the modulated data.

[0130] In some implementations, the selected FEC code may be a low density parity check (LDPC) code or a binary convolutional (BCC) code. In some implementations, the target data rate may be 1 Mbps or 1.5 Mbps, and the range of the target data rate may be 20% of the target data rate. For example, Table 1 and Table 2 show that a resulting data rate is within 20% of the target data rate (e.g., 1 Mbps or 1.5 Mbps).

[0131] In some implementations, the FEC code may be an LDPC code with the first code rate (e.g., 1/2) that is greater than the selected code rate (e.g., 1/3). In encoding the data, the one or more processors may be configured to receive a first set of information bits (e.g., information bits **1301** in FIG. **13**). The one or more processors may be configured to concatenate the first set of information bits with a set of information bits (e.g., set of information bits **1302**) to generate a second set of information bits (e.g., set of information bits **1310**). The one or more processors may be configured to encode, by the FEC encoder using the LDPC code with the first code rate (e.g., 1/2), the second set of information bits **1310**, to generate parity data (e.g., parity data **1350**). The one or more processors may be configured to generate the encoded data by concatenating the first set of information bit **1301** and the parity data **1350** to achieve the selected code rate.

[0132] In some implementations, the FEC code may be an LDPC code with the first code rate (e.g., 1/2) that is greater than the selected code rate (e.g., 1/3). In encoding the data, the one or more processors may be configured to receive a set of information bits. The one or more processors may be configured to encode, by the FEC encoder, the set of information bits using the LDPC code with the first code rate to generate encoded bits and parity data. The one or more processors may be configured to puncture one or more bits from the parity data to generate punctured parity data. The one or more processors may be configured to generate the encoded data by concatenating the encoded bits and the punctured parity data. For example, Referring to FIG. **7**B, the system may discard some parity bits (e.g., "punctured bits" **741**) after encoding. In this manner, the system can increases the effective code rate and degrade a coding gain

[0133] In some implementations, the FEC code may be a first BCC code with the first code rate (e.g., 2/3) that is greater than the selected code rate (e.g., 1/3). In encoding the data, the one or more processors may be configured to concatenate the first BCC code with the first code rate with a repetition code (e.g., a native 2/3 BCC code concatenated with a 1/2 repetition code; see Table 2) to generate a second BCC code that corresponds to the selected code rate (e.g., 1/3). The one or more processors may be configured to encode, by the FEC encoder, the data using the second BCC code, to generate the encoded data that corresponds to the selected code rate.

[0134] FIG. **17** is a flow diagram showing a process **1700** for obtaining a particular data rate using various schemes of encoding, modulation and/or transmitting, in accordance with an embodiment. In some implementations, the process **1700** for wireless communications over one or more channels is performed by one or more processors of a system (e.g. processor **2010** of communication system **105**, transmitter circuitry **120**, baseband circuitry **110**, encoder **130**). In other embodiments, the process **1700** is performed by other entities (e.g., a computing system other than the communication system **105**). In some implementations, the process **1700** includes more, fewer, or different steps than shown in FIG. **17**.

[0135] At step **1702**, the one or more processors of the system may identify a target data rate (e.g., 1 Mbps or 1.5 Mbps) for transmitting data over a channel with a frequency bandwidth (e.g., 20 MHz).

[0136] At step **1704**, based at least on the target data rate, the one or more processors may select a forward error correction (FEC) code, a code rate, a modulation scheme, and a number of resource units (RUs) within the frequency bandwidth, to transmit the data within a range of the target data rate. For example, based on the target data rate (e.g., 1 Mbps or 1.5 Mbps), the apparatus can select a code rate, a modulation scheme, and a number of RUs using Table 1 or Table 2. In some implementations, the target data rate may be 1 Mbps or 1.5 Mbps, and the range of the target data rate may be 20% of the target data rate. For example, Table 1 and Table 2 show that a resulting data rate is within 20% of the target data rate (e.g., 1 Mbps or 1.5 Mbps).

[0137] In some implementations, the method may include based at least on the target data rate, selecting a guard interval (GI) and a number of tones per RU, to transmit the data within a range of the target data rate. For example, based on the target data rate (e.g., 1 Mbps or 1.5 Mbps), the apparatus can select a GI, and a number of RUs using Table 1 or Table 2.

[0138] At step **1706**, the one or more processors may encode, by an FEC encoder, the data using the FEC code and the code rate to generate encoded data. In some implementations, the selected FEC code may be a low density parity check (LDPC) code or a binary convolutional (BCC) code. In some implementations, the FEC code may have a first code rate that is different from the selected code rate. The method may include encoding, by the FEC encoder, the data using the FEC code with the first code rate to generate encoded data that corresponds to the selected code rate. For example, referring to FIG. **13**, the LDPC encoder **1340** can encode data using an LDPC code with a code rate of 1/2 to generate encoded data that corresponds to the selected code rate of 1/3.

[0139] In some implementations, the FEC code may be an LDPC code, the selected code rate may be 1/3, and the first code rate may be ½ or 1/4 (see Table 1). In some implementations, the FEC code may be a BCC code, the selected code rate may be 1/3, and the first code rate may be 2/3 (see Table 2).

[0140] At step **1708**, the one or more processors may modulate the encoded data using the modulation scheme (e.g., BPSK or QPSK) to generate modulated data.

[0141] At step **1710**, a transmitter of the system may transmit the modulated data using the number of RUs (e.g., RU26, RU52, RU106, a combination of RU26 and RU52, a combination of RU26 and RU106 as shown in FIG. **10**).

[0142] References to "or" may be construed as inclusive so that any terms described using "or" may indicate any of a single, more than one, and all of the described terms. References to at least one of a conjunctive list of terms may be construed as an inclusive OR to indicate any of a single, more than one, and all of the described terms. For example, a reference to "at least one of 'A' and 'B'" can include only 'A', only 'B', as well as both 'A' and 'B'. Such references used in conjunction with "comprising" or other open terminology can include additional items.

[0143] It should be noted that certain passages of this disclosure can reference terms such as "first" and "second" in connection with subsets of transmit spatial streams, sounding frames, response, and devices, for purposes of identifying or differentiating one from another or from others. These terms are not intended to merely relate entities (e.g., a first device and a second device) temporally or according to a sequence, although in some cases, these entities can include such a relationship. Nor do these terms limit the number of possible entities (e.g., STAs, APs, beamformers and/or beamformees) that can operate within a system or environment. It should be understood that the systems described above can provide multiple ones of any or each of those components and these components can be provided on either a standalone machine or, in some embodiments, on multiple machines in a distributed system. Further still, bit field positions can be changed and multibit words can be used. In addition, the systems and methods described above can be provided as one or more computer-readable programs or executable instructions embodied on or in one or more articles of manufacture, e.g., a floppy disk, a hard disk, a CD-ROM, a flash memory card, a PROM, a RAM, a ROM, or a magnetic tape. The programs can be implemented in any programming language, such as LISP, PERL, C, C++, C#, or in any byte code language such as JAVA. The software programs or executable instructions can be stored on or in one or more articles of manufacture as object code.

[0144] While the foregoing written description of the methods and systems enables one of ordinary skill to make and use embodiments thereof, those of ordinary skill will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The present methods and systems should therefore not be limited by the above described embodiments, methods, and examples, but by all embodiments and methods within the scope and spirit of the disclosure.

What is claimed:

1. An apparatus comprising:

a transmitter and one or more processors, wherein

the one or more processors are configured to:

identify a target data rate for transmitting data over a channel with a frequency bandwidth;

based at least on the target data rate, select a forward error correction (FEC) code, a code rate, a modulation scheme, and a number of resource units (RUs) within the frequency bandwidth, to transmit the data within a range of the target data rate;

encode, by an FEC encoder, the data using the FEC code and the code rate to generate encoded data; and

modulate the encoded data using the modulation scheme to generate modulated data, and

the transmitter is configured to transmit the modulated data using the number of RUs.

2. The apparatus according to claim **1**, wherein

the selected FEC code is a low density parity check (LDPC) code or a binary convolutional (BCC) code.

3. The apparatus according to claim **1**, wherein the one or more processors are further configured to:

based at least on the target data rate, select a guard interval (GI) and a number of tones per RU, to transmit the data within a range of the target data rate.

4. The apparatus according to claim **1**, wherein

the target data rate is 1 Mbps or 1.5 Mbps, and

the range of the target data rate is 20% of the target data rate.

5. The apparatus according to claim **1**, wherein

the FEC code has a first code rate that is different from the selected code rate,

the one or more processors are further configured to:

encode, by the FEC encoder, the data using the FEC code with the first code rate to generate encoded data that corresponds to the selected code rate.

6. The apparatus according to claim **1**, wherein

the FEC code is an LDPC code,

the selected code rate is ⅓, and

the first code rate is ½ or ¼.

7. The apparatus according to claim **1**, wherein

the FEC code is a BCC code,

the selected code rate is ⅓, and

the first code rate is ⅔.

8. A method, comprising:

identifying, by one or more processors, a target data rate for transmitting data over a channel with a frequency bandwidth;

based at least on the target data rate, selecting, by the one or more processors, a forward error correction (FEC) code, a code rate, a modulation scheme, and a number of resource units (RUs) within the frequency bandwidth, to transmit the data within a range of the target data rate;

encoding, by an FEC encoder, the data using the FEC code and the code rate to generate encoded data; and

modulating by the one or more processors, the encoded data using the modulation scheme to generate modulated data, and

transmitting, by the transmitter, the modulated data using the number of RUs.

9. The method according to claim **8**, wherein

the selected FEC code is a low density parity check (LDPC) code or a binary convolutional (BCC) code.

10. The method according to claim **8**, further comprising:

based at least on the target data rate, selecting a guard interval (GI) and a number of tones per RU, to transmit the data within a range of the target data rate.

11. The method according to claim **8**, wherein

the target data rate is 1 Mbps or 1.5 Mbps, and

the range of the target data rate is 20% of the target data rate.

12. The method according to claim **8**, wherein

the FEC code has a first code rate that is different from the selected code rate,

the method comprises:

encoding, by the FEC encoder, the data using the FEC code with the first code rate to generate encoded data that corresponds to the selected code rate.

13. The method according to claim **8**, wherein

the FEC code is an LDPC code,

the selected code rate is ⅓, and

the first code rate is ½ or ¼.

14. The method according to claim **8**, wherein

the FEC code is a BCC code,

the selected code rate is ⅓, and

the first code rate is ⅔.

15. An apparatus comprising:
a transmitter and one or more processors, wherein
the one or more processors are configured to:
identify a target data rate for transmitting data;
based at least on the target data rate, select a modulation scheme and a code rate to transmit the data within a range of the target data rate;
identify, based at least on the selected code rate, a forward error correction (FEC) code with a first code rate that is different from the selected code rate;
encode, by an FEC encoder, the data using the FEC code with first code rate to generate encoded data that corresponds to the selected code rate; and
modulate the encoded data using the modulation scheme to generate modulated data, and
the transmitter is configured to transmit the modulated data.

16. The apparatus according to claim **1**, wherein
the selected FEC code is a low density parity check (LDPC) code or a binary convolutional (BCC) code.

17. The apparatus according to claim **1**, wherein
the target data rate is 1 Mbps or 1.5 Mbps, and
the range of the target data rate is 20% of the target data rate.

18. The apparatus according to claim **1**, wherein
the FEC code is an LDPC code with the first code rate that is greater than the selected code rate;
in encoding the data, the one or more processors are configured to:
receive a first set of information bits;
concatenate the first set of information bits with a set of information bits to generate a second set of information bits;
encode, by the FEC encoder using the LDPC code with the first code rate, the second set of information bits, to generate parity data; and
generate the encoded data by concatenating the first set of information bit and the parity data to achieve the selected code rate.

19. The apparatus according to claim **1**, wherein
the FEC code is an LDPC code with the first code rate that is greater than the selected code rate;
in encoding the data, the one or more processors are configured to:
receive a set of information bits;
encode, by the FEC encoder, the set of information bits using the LDPC code with the first code rate to generate encoded bits and parity data;
puncture one or more bits from the parity data to generate punctured parity data; and
generate the encoded data by concatenating the encoded bits and the punctured parity data.

20. The apparatus according to claim **1**, wherein
the FEC code is a first BCC code with the first code rate that is greater than the selected code rate;
in encoding the data, the one or more processors are configured to:
concatenate the first BCC code with the first code rate with a repetition code to generate a second BCC code that corresponds to the selected code rate; and
encode, by the FEC encoder, the data using the second BCC code, to generate the encoded data that corresponds to the selected code rate.

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